TEXTBOOK · FALL 2026

Math I
Weeks 1–4

All questions and all formulas from every course file and the assigned book sections — number systems, notation, complex numbers; functions; limits and limit laws, standard limits, indeterminate forms, asymptotics and the Landau symbols — each problem with a complete step-by-step solution.
Compiled for independent study and Test 1 preparation

Tashkent Technological University · Fall 2026
Sources: course notes, lectures, ISA, practical sessions;
OpenStax Calculus 1

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1 Course Map and How to Use This Document

This document is a complete standalone guide for the first four weeks of the **Math I** course in the first semester (Fall 2026). It was manually compiled from each relevant source in the course-files/ folder of the course repository: weekly notes (Notes), lecture presentations, practical exercise sheets, independent study assignments (ISA), and designated sections of the OpenStax Calculus Volume 1 textbook. Problems are reproduced verbatim; solutions are complete, step-by-step, with verifications.

Week	Math I
1	Number sets N ⊆ Z ⊆ Q ⊆ R ⊆ C; positional systems; scientific and engineering notation; complex numbers
2	Functions: classes, domain/range, symmetry, composition, inverse functions; exponential and logarithmic functions (§1.2, §1.5)
3	Limits: intuitive concept, one-sided and infinite limits, asymptotes, precise ε–δ definition (§2.2, §2.5)
4	Limits II: limit laws, standard limits, indeterminate forms, squeeze theorem, asymptotic equivalence, Landau symbols O and o (§2.3)

Notation. Powers and indices are written using ^ and _: x^2, a_{11}, $f^{(-1)}$. Sets are written conventionally: N ⊆ Z ⊆ Q ⊆ R ⊆ C. The derivative is denoted f′(x) or dy/dx.

2 Week 1 — Number systems, notation, complex numbers (Math I)

2.1 Math I: Formula catalog for week 1

2.1.1 Number sets and positional systems

The inclusion of standard number sets is the first structural fact of the course, and every subsequent classification exercise relies on it. Rational numbers are exactly fractions with integer numerator and denominator, while irrational numbers are real numbers that are not rational. N ⊆ Z ⊆ Q ⊆ R ⊆ C, R \ Q = {x ∈ R : x ∉ Q} (irrational) (1) q ∈ Q ⇐⇒ q = a

$$a \in Z,\ b \in Z,\ b \neq 0,\ d(a, b) = |a - b| \text{ (distance in R) (2)}$$

A positional numeral system with base b represents a number through powers of the base; the same number admits many representations, but its value does not depend on the representation. Conversion between bases relies on the expansion below (and on successive division for the reverse direction).

$$X_n\,(a_n a_{n-1} \ldots a_1 a_0)_b = \Sigma_{k=0}^{\ n}\, a_k\, b^k,\ 0 \le a_k \le b - 1, \text{ (3)}$$

$$13_{10} = 1101_2 = 15_8 = D_{16},\ A = 10,\ B = 11, \ldots, F = 15 \text{ (hex digits). (4)}$$

Roman numerals are non-positional: I = 1, V = 5, X = 10, L = 50, C = 100, so 13 = XIII; arithmetic in such systems is impractical, so computers use the binary system: electronics reliably distinguishes two states (low/high, off/on, 0/1).

2.1.2 Scientific and engineering notation

Scientific notation highlights the order of magnitude of a number in a coefficient with one non-zero digit before the decimal and a power of ten; engineering notation requires the exponent to be a multiple of three to align with SI prefixes.

$$\text{scientific: } z = a \times 10^{n},\ 1 \le |a| < 10,\ n \in Z,\ (5) \quad \text{engineering: } z = a \times 10^{3k},\ 1 \le |a| < 1000,\ k \in Z,\ (6)$$

$$a \times 10^{m-n} = a \times 10^{m-n}\ (7)$$

$(a \times 10^{m})(b \times 10^{n}) = ab \times 10^{m+n}$, $b \times 10^{m-n}$, (7)

$$(a \times 10^{n})^{m} = a^{m} \times 10^{nm}$$

add./sub.: first to common power of ten (8) SI prefixes used throughout the semester:

Power	10^{12}	10^{9}	10^{6}	10^{3}	10^{-3}	10^{-6}	10^{-9}	10^{-12}	10^{-15}
Prefix	**tera**								
(T)	**giga**								
(G)	**mega**								
(M)	**kilo**								
(k)	**milli**								
(m)	**micro**								
(µ)	**nano**								
(n)	**pico**								
(p)	**femto**								
(f)									

Case matters: 5 mW = 5×10^{-3} W, but 5 MW = 5×10^{6} W. Reference values used throughout the course: c = 2.99792458×10^{8} m/s; e = $1.602176634 \times 10^{-19}$ C; $N_A = 6.02214076 \times 10^{23}\ \text{mol}^{-1}$; h = $6.62607015 \times 10^{-34}$ J·s; 101325 Pa = 101.325 kPa.

2.1.3 Complex numbers

Week 1 concludes with complex numbers, which arise because the equation $x^2 + 1 = 0$ has no real solutions. The imaginary unit, equality rule, four arithmetic operations, and the complex conjugate are the working tools. √

$$i = \sqrt{-1},\ i^2 = -1,\ z = a + bi,\ Re(z) = a,\ Im(z) = b,\ (9)\ i^{n+4} = i^{n}\ (i^{0} = 1,\ i^{1} = i,\ i^{2} = -1,\ i^{3} = -i),$$

$a + bi = c + di \iff a = c$ and $b = d$, (10)

$$(z_1 + z_2) = (a + c) + (b + d)i,\ z_1 - z_2 = (a - c) + (b - d)i,\ (11)\ (a + bi)(c + di) = (ac - bd) + (ad + bc)i,$$
$$(z_1)/(z_2) = (z_1\ overline{z_2})/(|z_2|^2)\ (z_2 \neq 0),\ (12)\ \bar{z} = a - bi,\ z + \bar{z} = 2\ Re(z),\ z - \bar{z} = 2i\ Im(z),\ |z|^2 = a^2 + b^2 = z\bar{z},\ (13)\ |z| = sqrt\{a^2 + b^2\},\ |z_1 z_2| = |z_1|\ |z_2|,\ |(z_1)/(z_2)| = (|z_1|)/(|z_2|),\ (14)\ |z_1 + z_2| \le |z_1| + |z_2|$$
(triangle inequality) (15)

$\tan\theta = b$ $\arg z = \theta + 2k\pi$, $k \in Z$; $\text{Arg}(z) \in (-\pi, \pi]$ principal value, a (quadrant!) (16) $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$, $r = |z|$, $a = r\cos\theta$, $b = r\sin\theta$, (17) $e^{i\theta} = \cos\theta + i\sin\theta$ $e^{i\pi} + 1 = 0$, (Euler), (18)

$$z_1\ z_2 = r_1\ r_2\ e^{i(\theta_1 + \theta_2)},\ (z_1)/(z_2) = (r_1)/(r_2)\ e^{i(\theta_1 - \theta_2)},\ (18)$$

(19)

$$z_2\ r_2\ z_n = r_n\ e^{i(n\theta_1 + \theta_2)},\ z_2 = r_2\ e^{-i\theta_2}\ (19)$$

$\cos(n\theta) + i\sin(n\theta)$ (De Moivre's formula), (20)

$$z_k = r^{1/n}\ (\ cos((\theta + 2k\pi)/(n)) + isin((\theta + 2k\pi)/(n))\ ),\ k \in Z$$

cos + i sin, k = 0, . . . , n − 1, (21) n n

$$z^n = 1 \Rightarrow z_k = e^{2\pi ik/n},\ k = 0, \ldots, n-1 \text{ (n-th roots of 1), (22)}$$

(22) √

$$z = (-b \pm \text{sqrt}(b^2 - 4ac))/(2a)$$

(quadratic formula, also valid over C). (23) 2a The roots of the equation $z^n = w = re^{i\theta}$ are given by formula (21); the n-th roots of unity form a regular n-gon on the unit circle. Non-real roots of polynomials with real coefficients occur in conjugate pairs, and by the fundamental theorem of algebra, a polynomial of degree n has exactly n roots in C, counting multiplicity.

2.2 Math I: Week 1 Practical Session — questions and complete solutions

1. Classify each decimal as terminating, periodic, or non-periodic, and state whether the corresponding number is rational: (1) 0.375; (2) 0.272727 . . .; (3) 1.101001000100001 . . .; (4) 3.14159265 . . .; (5) 0.123123123 . . .

[Source: Week 1, practical session, exercise 1]

Solution. (1) 0.375 is finite, therefore rational: $0.375 = {}^{375} = {}_{8}^{3}$. (2) 1000 Block 27 repeats infinitely, therefore rational; for a purely periodic block ab $= {}_{99}^{ab}$, so $0.27 = 27_{99} = 3_{11}$. (3) The intervals between consecutive ones grow (1, 2, 3, . . . zeros), so the fraction will never become periodic or terminate: the number is irrational. (4) A finite number of digits shown with ellipsis does not define the number; if π = 3.14159265 . . . was intended, it is infinite and non-periodic, therefore irrational. (5) Block 123 repeats, therefore rational: $0.123 = {}^{123}_{999} = {}^{41}_{333}$. Principle: finite or eventually periodic decimal fractions are exactly the rational numbers.

2. Write each set in interval notation and represent it on the number line: (1) {x ∈ R : −3 ≤ x < 4}; (2) {x ∈ R : x ≠ 3}.

[Source: Week 1, practical session, exercise 2]

Solution. (1) The endpoint −3 is included (square bracket), 4 is excluded (parenthesis): [−3, 4); on the number line, this is shown as a filled point at −3 and an open point at 4, with the segment between them highlighted. (2) All real numbers except 3: (−∞, 3) ∪ (3, ∞); on the number line, the entire axis is filled with an open point at 3. Infinity is never included, so parentheses are always used.

3. Write each interval in inequality form: a. [−2, 5); b. (−∞, 4]; c. (1, ∞).

[Source: Week 1, practical session, exercise 3]

Solution. a. −2 ≤ x < 5 (left endpoint included, right endpoint excluded). b. x ≤ 4 (unbounded to the left, right endpoint included). c. x > 1 (strictly greater than, since the parenthesis excludes 1). √ √

4. Classify as rational or irrational: a. 3; b. 25; c. √ √ √ 50; i. (1 + 5)(1 − 5). √

[Source: Week 1, practical session, ex. 4]

Solution. a. 3 is not a perfect square, therefore √3 is irrational. b. √25 = 5 ∈ Z ⊆ Q, rational. c. √50 = 5√2; since √2 is irrational, 5√2 is irrational. i. Expanding: (1

+ √5)(1 − √5) = 1 − 5 = −4, rational. The product of conjugates always eliminates the radical.

5. Given $z_1 = 3 + 2i$ and $z_2 = 1 − 4i$; compute $z_1 + z_2$, $z_1 − z_2$, z_1z_2, z_1/z_2, writing each answer in the form a + bi.

[Source: Week 1, practical session, ex. 5]

Solution. Addition and subtraction act on the real and imaginary parts separately: $z_1 + z_2 = (3+1)+(2 − 4)i = 4 − 2i$; $z_1 − z_2 = (3 − 1)+(2+4)i = 2+6i$. In the product, we use $i^2 = −1$ and the rule (a + bi)(c + di) = (ac − bd) + (ad + bc)i: $z_1z_2 = (3 · 1 − 2 · (−4)) + (3 · (−4) + 2 · 1)i = (3 + 8) + (−12 + 2)i = 11 − 10i$. In the quotient, we multiply numerator and denominator by the conjugate of the denominator:

$$(z_1)/(z_2) = ((3 + 2i)(1 + 4i))/((1 − 4i)(1 + 4i)) = (3 + 12i + 2i + 8i^2)/(1 + 16) = ((3 − 8) + (12 + 2)i)/17 = (-5 + 14i)/17 = -5/17 + (14/17)i.$$

6. For the numbers 3 + 4i, −2 + 5i, 6i, −7, find the conjugate and modulus; verify $z\bar{z} = |z|^2$. √ $a^2 + b^2$. 3 + 4i: $\bar{z} = 3 − 4i$, $|z| = 5$,

[Source: Week 1, practical session, ex. 6]

Solution. For z = a + bi we have $\bar{z} = a − bi$ and |z| =

$$\sqrt{a^2 + b^2}$$

. −2 + 5i: $\bar{z} = −2 − 5i$, |z| = sqrt(4 + 25) = sqrt(29) □ and (3 − 4i)(3 + 4i) = 9 + 16 = 25 = 5^2. 29, and

$$6i: \bar{z} = −6i,\ |z| = 6, \text{ and } (−6i)(6i) = −36i^2 = 36.$$

(−2 − 5i)(−2 + 5i) = 4 + 25 = 29. −7: $\bar{z} = −7$ (real numbers are conjugates of themselves), |z| = 7, (−7)(−7) = 49. All four checks agree.

1

7. Express 1/(1 + i) in the form a + bi.

[Source: Week 1, practical session, ex. 7]

Solution. Multiply by the conjugate of the denominator:

$$1/(1 + i) = (1 − i)/((1 + i)(1 − i)) = (1 − i)/2 = 1/2 − i/2$$

8. Solve over C: $x^2 + 9 = 0$.

[Source: Week 1, practical session, ex. 8]

Solution. Rearranging, we get $x^2 = −9$, so x = ±√−9 = ±3i. Both solutions are non-real complex numbers; there are no real solutions because a square in R is never negative.

9. For z = 2 + 3i on the complex plane, determine the real part, imaginary part, modulus, and quadrant. What changes for 2 − 3i?

[Source: Week 1, practical session, ex. 9]

Solution. The point (2, 3) lies in the first quadrant: Re(z) = 2, Im(z) = 3, |z| =

$$\sqrt{(2^2 + 3^2)} = \sqrt{13}$$

. For z = 2 − 3i, the point (2, −3) is the reflection across the real axis: the real part and modulus do not change, the imaginary part changes sign, and the point moves to the fourth quadrant.

10. Write 1 + i in trigonometric form $z = r(\cos\theta + i\sin\theta)$ and exponential form $z = re^{i\theta}$, with θ in radians.

[Source: Week 1, practical session, ex. 10]

Solution. Here $r = |1 + i| = \sqrt{1 + 1} = \sqrt{2}$, both coordinates are positive, so $\theta = \arctan(1/1) = \pi/4$. Thus, $1 + i = \sqrt{2}(\cos\pi/4 + i\sin\pi/4) = \sqrt{2}\, e^{i\pi/4}$.

11. Using De Moivre's formula $[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)$, compute $(\sqrt{3} + i)^4$; verify algebraically. $|\sqrt{3} + i| = \sqrt{3} + 1 = 2$, $\theta = \arctan(1/\sqrt{3}) = \pi/6$.

[Source: Week 1, practical session, ex. 11]

Solution. First, convert: $r = |\sqrt{3} + i| = \sqrt{3} + 1 = 2$, $\theta = \arctan(1/\sqrt{3}) = \pi/6$. By De Moivre's,

$$(\sqrt{3} + i)^4 = 2^4(\cos(4\arctan(1/\sqrt{3})) + i\sin(4\arctan(1/\sqrt{3}))) = 16(\cos(4\pi/6) + i\sin(4\pi/6)) = 16(\cos(2\pi/3) + i\sin(2\pi/3)) = 16(-1/2 + \sqrt{3}/2\, i) = -8+8\sqrt{3}i.$$

Algebraic verification:

$$(\sqrt{3} + i)^2 = 3 + 2\sqrt{3}i + i^2 = 2 + 2\sqrt{3}i;$$

squaring again,

$$(2 + 2\sqrt{3}i)^2 = 4 + 8\sqrt{3}i + 12i^2 = -8 + 8\sqrt{3}i.$$

Both methods agree. 12. Find all solutions to $z^3 = -8$ and "plot them on the COMPLEX PLANE. [Source: Week 1, practical session, ex. 12] **Solution.** [The number -8 = 8e^(iπ), cube roots have modulus 8^(1/3) = 2 and arguments (π + 2πk)/3, k = 0, 1,2: z_0 = 2e^(iπ/3) = 1+√3i, z_1 = 2e^(iπ) = -2, z_2 = 2e^(i5π/3) = 1-√3i.]

12. Find all solutions to $z^3 = -8$ and plot them on the complex plane.

[Source: Week 1, practical session, exercise 12]

Solution. We write $-8 = 8e^{i(\pi+2k\pi)}$. The cube roots have modulus $8^{1/3} = 2$ and arguments $\pi+2k\pi$, $k = 0, 1, 2$:

$$z_0 = 2e^{i\pi/3} = 1 + sqrt(3)\, i,\ z_1 = 2e^{i\pi} = -2,\ z_2 = 2e^{i5\pi/3} = 1 - sqrt(3)\, i.$$

Geometrically, these are three equally spaced points on a circle of radius 2 centered at the origin (angles 60°, 180°, 300°); note that $z_1 = -2$ is the only real solution, which is consistent with $(-2)^3 = -8$.

13. For each number, specify the smallest set from N, Z, Q, R, C to which it belongs: 0, −7, $\frac{5}{8}$, 3.25, 16, 7, $\sqrt{-9}$, π, π^2, i, 4 − 3i.

[Source: Week 1, ISA, assignment 2]

Solution. $0 \in \mathbb{N}$ (by course convention) ⇒ smallest set N. $-7 \in \mathbb{Z}$. $\frac{5}{8}$ and $3.25 = \frac{13}{4}$ lie in Q (finite decimal fractions). $16 = 4^2 \in \mathbb{N}$. √7 is irrational (7 is not a perfect square) ⇒ R. $-\sqrt{9} = -3 \in \mathbb{Z}$. π and π^2 are irrational ⇒ R (squaring does not make an irrational number rational). i and 4 − 3i are non-real ⇒ only C.

14. True or false? Correct each false statement: (1) $\mathbb{N} \subseteq \mathbb{Z}$; (2) $\mathbb{Z} \subseteq \mathbb{Q}$; (3) $\mathbb{R} \subseteq \mathbb{Q}$; (4) $\mathbb{Q} \subseteq \mathbb{R}$; (5) $\mathbb{R} \subseteq \mathbb{C}$; (6) every integer is natural; (7) every rational number is real; (8) every complex number is real; (9) every real number can be written as a complex number; (10) a number cannot be both rational and real.

[Source: Week 1, ISA, assignment 3]

Solution. (1) True. (2) True. (3) False — for example, $\sqrt{2} \in \mathbb{R}$, but $\sqrt{2} \notin \mathbb{Q}$; the correct inclusion is $\mathbb{Q} \subseteq \mathbb{R}$. (4) True. (5) True. (6) False — negative integers, for example -3, are not natural. (7) True. (8) False — for example, i is complex but not real. (9) True — $a = a + 0i$. (10) False — every rational number is real (statement (4)), so rational numbers are both.

15. Draw a diagram of nested sets $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ and place $\sqrt{5}$, -2, $\frac{2}{3}$, 2, π, $-4i$, $3 + 2i$ in the correct regions.

[Source: Week 1, ISA, assignment 4]

Solution. Draw five nested ovals labeled $\mathbb{N}$ (innermost) to $\mathbb{C}$ (outermost). Placement: $5 \in \mathbb{N}$; $-2 \in \mathbb{Z} \setminus \mathbb{N}$; $\frac{2}{3} \in \mathbb{Q} \setminus \mathbb{Z}$; 2 and $\pi \in \mathbb{R} \setminus \mathbb{Q}$; $-4i$ and $3 + 2i \in \mathbb{C} \setminus \mathbb{R}$. The diagram visually shows that each new set only adds numbers, removing nothing.

16. Arrange in increasing order: $-\sqrt{5}$, -2, $-\frac{3}{2}$, 0, 2, $\frac{3}{2}$, $\frac{\pi}{2}$; justify non-obvious comparisons.

$$(2/3)^2 = 4/9 < (\pi/2)^2$$

[Source: Week 1, ISA, assignment 5]

Solution. Squares give the order for positive numbers: $(2/2)^2 = (\frac{3}{2})^2$; since $\pi/2 \approx 1.5708$ and $3/2 = 1.5$, we have $3/2 < \pi/2$. For negative numbers, the order is reversed: $-\sqrt{5} \approx -2.236 < -2 < -3/2$. In total,

$$-sqrt(5) < -2 < -3/2 < 0 < 2 < 3/2 < \pi/2$$

17. Write each set in interval notation: (1) $-3 \le x < 4$; (2) $x > 2$; (3) $x \le -1$; (4) $-2 < x \le 5$; (5) $x \ne 3$; then graph each on the number line.

[Source: Week 1, ISA, assignment 6]

Solution. (1) $[-3, 4)$. (2) $(2, \infty)$. (3) $(-\infty, -1]$. (4) $(-2, 5]$. (5) $(-\infty, 3) \cup (3, \infty)$. On the number line: filled points where the endpoint is included ($\le$, $\ge$), open points where it is not ($<$, $>$), parentheses with infinities, and two rays for the union in (5).

18. Write each interval as an inequality: a. $[-2, 5)$; b. $(-\infty, 4]$; c. $(1, \infty)$; d. $(-3, 0] \cup [2, 6)$; e. $\mathbb{R} \setminus [-1, 1]$.

[Source: Week 1, ISA, assignment 7]

Solution. a. $-2 \le x < 5$. b. $x \le 4$. c. $x > 1$. d. $-3 < x \le 0$ or $2 \le x < 6$. e. $x < -1$ or $x > 1$; equivalently $|x| > 1$.

19. With $d(a, b) = |a - b|$: compute $d(3, 8)$, $d(-2, 5)$, $d(-4, -9)$, $d(2, 8)$; check $d(a, c) \le d(a, b) + d(b, c)$ for $a = -3$, $b = 2$, $c = 7$.

$$d(2, 8) = | 2-8 | / 2 = 2$$

[Source: Week 1, ISA, assignment 8]

Solution. $d(3, 8) = |3-8| = 5$; $d(-2, 5) = 7$; $d(-4, -9) = 5$; $d(2, 8) = | 2-8 | / 2 = 2$. Checking the triangle inequality: $d(-3, 7) = 10$, $d(-3, 2) + d(2, 7) = 5 + 5 = 10$, and indeed $10 \le 10$ (equality, since the points lie in order on a straight line). The formula $|a - b|$ is the one-dimensional case of the norm from (28).

20. Determine whether each number is rational or irrational; justify: a. $\sqrt{49}$; b. $\sqrt{18}$; c. $\sqrt{3} + \sqrt{2}$; d. $\sqrt{2} \cdot \sqrt{5}$; e. $\sqrt{2} + \sqrt{8}$; f. $(\sqrt{3})^2$; g. $(1 + \sqrt{2})(1 - \sqrt{2})$; h. $\sqrt{12}$.

[Source: Week 1, ISA, assignment 9]

Solution. a. 49 = 7 is rational. b. 18 = 3 2 is irrational. c. √ rational + irrational = irrational: 3 + 2 is irrational.

rational + irrational = irrational: 3 + sqrt(2) irrational. n 9./— awRANHATTTATIA WLAN a f/9 | .fe ~— .f6 | 9.f/9) ~— 292./9 awrRNnATTIATTAULAUaID

d. 2 5 is irrational. e. 2 + 8 = 2 + 2 2 = 3 2 is irrational. √ 3)^2 = 3 is rational. f. (g. 1 − 2 = −1 is rational (product of √ √ conjugates). h. 12 = 2 3 is irrational. Practical rule: √ simplification reveals nature; products of conjugates of the form (a+ b)(a− √ b) are always rational.

21. Given $z_1 = 3 + 2i$, $z_2 = 1 - 4i$; compute $z_1 + z_2$, $z_1 - z_2$, $z_1 z_2$, z_1/z_2.

[Source: Week 1, ISA, assignment 10]

Solution. Same as exercise 5 of the practical session above: $z_1 + z_2 = 4 - 2i$, $z_1 - z_2 = 2 + 6i$, $z_1 z_2 = 11 - 10i$, and $z^1 = -17\,5 + 14\,17\,i$.

$$z_2 = 1 - 4i$$

22. Compute i^2, i^3, i^4, i^5, i^{18}, i^{27}, i^{100}; describe the repeating pattern and the method based on dividing the exponent by 4.

[Source: Week 1, ISA, assignment 11]

Solution. $i^2 = -1$; $i^3 = i^2 i = -i$; $i^4 = (i^2)^2 = 1$; $i^5 = i$. Powers are cyclic with period 4: 1, i, −1, −i, . . . For a large exponent, divide by 4 with remainder r: $i^n = i^r$. Then $18 = 4 \cdot 4 + 2 \Rightarrow i^{18} = i^2 = -1$; $27 = 4 \cdot 6 + 3 \Rightarrow i^{27} = i^3 = -i$; $100 = 4 \cdot 25 \Rightarrow i^{100} = i^0 = 1$.

23. For 3 + 4i, −2 + 5i, 6i, −7: find the conjugate and modulus, verify $zz = |z|^2$.

[Source: Week 1, ISA, assignment 12]

Solution. Same calculations as in exercise 6 of the practical session: √ conjugates 3 − 4i, −2 − 5i, −6i, −7; moduli 5, 29, 6, 7; in each case $zz = a^2 + b^2 = |z|^2$ holds exactly.

24. Express in the form a + bi: a. $\frac{1}{1+i}$; b. $\frac{3+2i}{2-i}$; c. $\frac{1+3i}{4-i}$; d. $\frac{5}{2-i}$.

[Source: Week 1, ISA, assignment 13]

Solution. In each case, multiply numerator and denominator by the conjugate of the denominator. a.

$$(1+i)/(1-i) = (1+i)^2/((1-i)(1+i)) = (1+2i+i^2)/(1-i^2) = (1+2i-1)/(1-(-1)) = (2i)/2 = i$$

= i. b.

$$(3+2i)/(2-i) = (3+2i)(2+i)/((2-i)(2+i)) = (6+3i+4i+2i^2)/(4-i^2) = (6+7i-2)/(4-(-1)) = (4+7i)/5$$

= 4/5 + 7/5 i. c.

$$(1+3i)/(4-i) = (1+3i)(4+i)/((4-i)(4+i)) = (4+i+12i+3i^2)/(16-i^2) = (4+13i-3)/(16-(-1)) = (1+13i)/17$$

= 1/17 + 13/17 i. d.

$$5i/(2-i) = 5i(2+i)/((2-i)(2+i)) = (10i+5i^2)/(4-i^2) = (10i-5)/(4-(-1)) = (-5+10i)/5$$

= -1 + 2i.

25. Solve over C, indicating which solutions are real: a. $x^2 + 9 = 0$; b. $x^2 - 4x + 13 = 0$; c. $2x^2 + 4x + 10 = 0$; d. $x^4 - 16 = 0$.

[Source: Week 1, ISA, assignment 14]

Solution. a. x = ±3i (non-real). b. Discriminant Δ= $(-4)^2 - 4 \cdot 13 = \sqrt{16 - 52} = -36$, Δ= 6i; x = $\frac{4 \pm 6i}{2} = 2 \pm 3i$ (non-real). c. Divide by 2:

$$x = (-4 \pm \sqrt{4^2 - 4 \cdot 2 \cdot 10})/(2 \cdot 2) = (-4 \pm \sqrt{-72})/4 = (-4 \pm 6i\sqrt{2})/4 = -1 \pm (3i\sqrt{2})/2.$$

$x^2 + 2x + 5 = 0$, Δ= 4 − 20 = −16, Δ= 4i; x = $\frac{-2 \pm 4i}{2} = -1 \pm 2i$ (non-real).

d. $x^4 = 16$ gives real solutions x = ±2; over C the complete set is the four fourth roots of 16: 2, −2, 2i, −2i (all solutions $(x^2 - 4)(x^2 + 4) = 0$).

26. For 2+3i, −1+4i, −3 − 2i, 4 − i: find the real part, imaginary part, modulus, quadrant. √ √

[Source: Week 1, ISA, assignment 15]

Solution. 2 + 3i: (2, 3), modulus 13, quadrant I. −1 + 4i: (−1, 4), modulus 17, √ √ quadrant II. −3 − 2i: (−3, −2), modulus 13, quadrant III. 4 − i: (4, −1), modulus 17, quadrant IV. Note that moduli pair up, since conjugation preserves $a^2 + b^2$.

27. Write in trigonometric and exponential forms (θ in radians): (1) √ 1 + i; (2) −1 + i; (3) − 3 − i; (4) 2i; (5) −4. √ √ √ √

[Source: Week 1, ISA, assignment 16]

Solution. (1) r = 2, θ = π/4: $2e^{i\pi/4}$. (2) r = 2, quadrant II: θ = 3π/4: $2e^{i3\pi/4}$. (3) r = √3 + 1 = 2, quadrant III: reference angle arctan(1/√ 3) = π/6, so θ = π + π/6 = 7π/6: $2e^{i7\pi/6}$. (4) r = 2, θ = π/2: $2e^{i\pi/2}$. (5) r = 4, θ = π: $4e^{i\pi}$. Each angle is taken from tan θ = b/a with adjustment for quadrant, as warned in (16). √ √

28. Using De Moivre's formula, calculate (1) $(1+ i)^6$; (2) $(3+ i)^4$; (3) $(-1+ 3 i)^5$; check one result algebraically. √ √

$$(1 + i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8 e^{i3\pi/2} = 8(0 - i) = -8i.$$

[Source: Week 1, ISA, assignment 17]

Solution. (1) 1 + i = √2 $e^{i\pi/4}$. Then $(1 + i)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8 e^{i3\pi/2} = 8(0 - i) = -8i$. (2) From exercise 11 of the practical session: $(3 + i)^4$ = -8 + 8sqrt(3)i. (3) z = −1+ 3i: modulus |z| = √1 + 3 = 2; quadrant II, reference angle π/3, so θ = 2π/3: z = $2e^{i2\pi/3}$. Then $z^5 = 2^5 e^{i10\pi/3} = 32e^{i(10\pi/3 - 2\pi)} = 32e^{i4\pi/3} = 32(-1/2 - i\sqrt{3}/2) = -16 - 16\sqrt{3}\, i$. Verification (1) algebraically: $(1 + i)^2 = 2i$, $(1 + i)^4 = (2i)^2 = -4$, $(1 + i)^6 = -4 \cdot 2i = -8i$. Matches.

29. Find all solutions and plot them on the complex plane: a. $z^3 = 1$; b. $z^4 = 1$; c. $z^3 = -8$. Explain why the n-th roots are equidistant on a circle. √ √

[Source: Week 1, ISA, assignment 18]

Solution. a. By (22): $z_k = e^{2\pi ik/3}$, k = 0, 1, 2: 1, $-\frac{1}{2} + \frac{\sqrt{3}}{2}$ i, $-\frac{1}{2} - \frac{\sqrt{3}}{2}$ i. b. $z_k = e^{2\pi ik/4}$: 1, i, −1, −i. c. From exercise 12 of the practical session: √ √ 1+ 3i, −2, 1 − 3i. In all cases, the roots have the same modulus $r^{1/n}$ (thus lie on the same circle), and their arguments differ by exactly $\frac{2\pi}{n}$ (a full turn 2π is divided into n equal arcs) — that's why they form the vertices of a regular n-gon.

30. The electric charge e = 0.000000000000000000001602176634 C. Write it in scientific and engineering notation.

[Source: Week 1, ISA, assignment 19]

Solution. We count 18 digits so that the first non-zero digit is before the decimal point: e = 1.602176634 × 10^{-19} C (scientific). In engineering notation, the exponent must be a multiple of 3, so we shift by one digit: e = 160.2176634 × 10^{-21} C = 160.2176634 zC (zeptocoulombs); in practice, they write 160.2176634 × 10^{-21} C, since below 10^{-24} there is only zepto (10^{-21}) among the standard SI prefixes.

31. Fill in the table of SI prefixes: powers 10^{12}, 10^{6}, 10^{3}, 10^{-6}, 10^{-9}, 10^{-12} with prefixes/notations.

[Source: Week 1, ISA, assignment 20]

Solution. 10^{12} tera (T); 10^{6} mega (M); 10^{3} kilo (k); 10^{-6} micro (µ); 10^{-9} nano (n); 10^{-12} pico (p). Together with giga (G), milli (m), femto (f), atto (a), this table forms the complete SI ladder.

32. Week 1 summary, part 2, in-class problems: (1) write 0.00000824 in scientific and engineering notation; (2) write 7.35 × 10^{8} Hz in engineering notation; (3) write 0.0047 F with an SI prefix.

[Source: Week 1 summary, part 2]

Solution. (1) 0.00000824 = 8.24 × 10^{-6} (scientific); since −6 is a multiple of 3, this is already in engineering notation, for example 8.24 µA if the unit is ampere. (2) 10^{8} is not a multiple of 3; we rewrite 7.35 × 10^{8} = 735 × 10^{6} = 735 MHz. (3) 0.0047 = 4.7 × 10^{-3} = 4.7 mF.

33. Week 1 summary, part 3, in-class problems: (1) draw a diagram of nested sets N ⊂ Z ⊂ Q ⊂ R ⊂ C and place √ 2026, −9, $_{3}^{2}$, 10, π, −5i, −2 + 2i; (2) rational or irrational: a. √ √ √ √ √ √ √ 49, b. 18, c. 3 + 2, d. 2 5, e. 2 + 8, f. $(5)^{2}$. √

[Source: Week 1 summary, part 3]

Solution. (1) 2026 ∈ N; −9 ∈ Z \ N; $_{3}^{2}$ ∈ Q \ Z; 10 and π ∈ R \ Q; −5i and −2 + 2i ∈ √ C \ R. (2) a. 7, rational. b. 3 2, irrational. c. irrational. d. √ irrational. e. 3 2, irrational. f. 5, rational — same logic as in assignment 9, which this exercise repeats with different numbers.

34. Week 1 summary, part 3, in-class problems: given $z_1 = 2 + 3i$, $z_2 = 1 − 5i$; calculate $z_1 + z_2$, $z_1 − z_2$, $z_1 z_2$, z_1/z_2.

[Source: Week 1 summary, part 3]

Solution. $z_1 + z_2 = 3 − 2i$; $z_1 − z_2 = 1+ 8i$; product: $(2 \cdot 1 − 3 \cdot (−5))+(2 \cdot (−5)+ 3 \cdot 1)i = (2 + 15)+ (−10 + 3)i = 17 − 7i$; quotient:

$$(z_1)/(z_2) = ((2 + 3i)(1 + 5i))/((1 - 5i)(1 + 5i)) = (2 + 10i + 3i + 15i^2)/(1 - (5i)^2) = (2 + 13i - 15)/(1 + 25) = (-13 + 13i)/26 = -1/2 + 1/2\, i.$$

35. Explain why $1 + i\sqrt{3} = 2\cos\frac{\pi}{3} + i \sin \pi = 2e^{i\pi/3}$. 3

$$|1 + i\sqrt{3}| = sqrt(1^2 + (\sqrt{3})^2) = sqrt(1 + 3) = 2.$$

[Source: Week 1, notes, part 3]

Solution. The modulus is $|1 + i\sqrt{3}|$. The point $(1, \sqrt{3})$ lies in the first quadrant, and $\tan \theta = \sqrt{3}$, from which $\theta = \pi/3$. Substituting $r = 2$, $\theta = \pi/3$ into the polar form (17), and then into Euler's formula (18), we obtain both equalities.

3 Week 2 — Functions (Math I)

3.1 Math I: Formula catalog for week 2

3.1.1 Functions: definition, classes, symmetry, composition,

inverse functions A function assigns to each x in the domain exactly one value f(x); the vertical line test identifies relations that do not satisfy this. The natural domain is the largest set of real inputs for which the formula makes sense; three standard obstacles: division by zero, even roots of negative numbers, and logarithms of non-positive numbers. f : D → R, x 7→ f(x); D = domain, range f = {f(x) : x ∈ D}, (31) p 1 g(x) : g(x) ≠ 0, g(x) : g(x) ≥ 0, $\log_b$ g(x) : g(x) > 0. (32) Even and odd functions reflect symmetry with respect to the y-axis and origin respectively; composition chains functions; inverses exist only for bijective functions (horizontal line test) and are found by solving y = f(x) for x.

$$\textit{even: } f(-x) = f(x), \textit{ odd: } f(-x) = -f(x),$$

(33) □ □

$$(f \circ g)(x) = f(g(x)),\ (f \circ g) \neq (g \circ f) \textit{ in general, (34)}$$

f is injective ⇐⇒ $f(x_1) = f(x_2) \Longrightarrow x_1 = x_2$, (35) □ □

$$f^{-1}(f(x)) = x,\ f(f^{-1}(y)) = y, \textit{ (36)}$$

and (36) f^{-1}: graph = mirror image of f's graph with respect to y = x. (37) Classes of functions from §1.2 (OpenStax): linear, polynomial, rational, root (algebraic), exponential and logarithmic, trigonometric (transcendental). Every polynomial is smooth everywhere, and every rational function is defined where its denominator is nonzero.

3.1.2 Exponential and logarithmic functions (§1.5,

hyperbolic functions excluded) Exponential and logarithm are inverses; laws of exponents and logarithms turn multiplication into addition and are used in every growth/decay model.

$$b^x \cdot b^y = b^{x+y},\ (b^x)^y = b^{xy},\ b^{-x} = 1/b^x,\ b^0 = 1,\ b > 0,\ b^1 = b,\ (b^x)_y = b^{xy}, \textit{ (38)}$$

bx, (38) $\log_b x = y \Longleftrightarrow b^y = x$,

$$x > 0,\ b > 0,\ b \neq 1, \textit{ (39) } x$$

$\log_b x^n = n \log_b x$, $\log_b(xy) = \log_b x + \log_b y$, $\log_b y = \log_b x - \log_b y$, (40) $\log_b x = \ln x\ b^x = e^{x \ln b}$, (41) (change of base), $\ln b\ e^{\ln x} = x$, $\ln 1 = 0$, $\ln e = 1$. (42) Growth and decay models compare $A(t) = A_0 a^t$ with $A_0 e^{kt}$ through the growth rate constant; doubling time is a benchmark quantity. $A(t) = A_0 a^t = A_0 e^{kt}$, k = ln a, a > 1 : growth; 0 < a < 1 : decay, (43) $k = \ln 2\ T_d = \ln 2\ N(t) = N_0\ 2^{t/Td} = N_0 e^{kt}$, , k. (44)

$$N(t) = N_0\, e^{-\lambda t}\ (\lambda = (\ln 2)/\tau)$$

Degrees and radians are converted via π rad = 180°: $\theta_{rad} = \theta_{deg} \cdot \frac{\pi}{180}$.

3.2 Math I: Week 2 workshop (functions) — questions and full solutions

1. Determine whether each relation defines y as a function of x: a. y = 3x −

$$x^2 + y^2 = 9.$$

2; b. Where appropriate, solve for y and apply the vertical line test.

[Source: Week 2, workshop, exercise 1]

Solution. a. y = 3x − 2 gives exactly one y for each x: a function (a line). √ b. Solving gives y = ± 9 − x^2: most x (for example x = 0) give two y values = ±3, so a vertical line intersects the circle twice; not a function.

2. Let f(x) = $2x^2$ − 3x + 1. Calculate and simplify f(0), f(−2), f(a), f(a + h) and f(a+h)−f(a) , h ≠ 0. h

[Source: Week 2, workshop, exercise 2]

Solution. f(0) = 1; f(−2) = 2(4) + 6 + 1 = 15; f(a) = $2a^2$ − 3a + 1; f(a + h) = 2(a + h)^2 − 3(a + h) + 1 = $2a^2$ + 4ah + $2h^2$ − 3a − 3h + 1. Difference quotient:

$$(f(a + h) − f(a))/h = (4ah + 2h^2 − 3h)/h = 4a + 2h − 3.$$

This expression anticipates the derivative of weeks 3–4 (as h → 0 it approaches 4a − 3).

$$:_{x2+9}; b.\ g(x) = \sqrt{5} − 2x.$$

3. Find the natural domain: a. f (x) = $^{x-1}$ Write in interval notation. The denominator x^2 + 9 ≥ 9 > 0 never equals zero,

[Source: Week 2, practical session, ex. 3]

Solution. a. therefore the domain is all R = (−∞, ∞). b. We need 5 − 2x ≥ 0 ⟺ x ≤ $_2^5$: domain (−∞, $_2^5$].

4. Let f (x) = 2x − 3, g(x) = x^2 + 1. Find (f ◦ g)(x) and (g ◦ f)(x); are they equal?

[Source: Week 2, practical session, ex. 4]

Solution. (f ◦ g)(x) = f (g(x)) = 2(x^2 + 1) − 3 = $2x^2$ − 1; (g ◦ f)(x) = g(2x − 3) = (2x − 3)^2 + 1 = $4x^2$ − 12x + 10. They are different (for example, at x = 0: −1 vs 10), which confirms: composition is generally non-commutative — squaring then shifting is not the same as shifting then squaring.

5. Determine whether each function is even, odd, or of general form: |x| + 1 and $_{x2+1}^{x}$; check algebraically and graphically.

[Source: Week 2, practical session, ex. 5]

Solution. f (x) = |x| + 1: f (−x) = | − x| + 1 = |x| + 1 = f (x) — even; the graph is symmetric with respect to the y-axis. g(x) = $_{x2+1}^{x}$: g(−x) = $\frac{-x}{2+1}$ = −g(x) — odd; the graph is symmetric with respect to 180◦ rotation about the origin.

6. Classify each function as linear, polynomial, rational, algebraic, or transcendental (a function can √ $^{x+1}$ x^2 + 1. belong to more than one category): $_{x2+4}$ and

[Source: Week 2, practical session, ex. 6]

Solution. $^{x+1}$ — a ratio of polynomials, hence rational (and

sqrt(x^2+4)

x^2 + 1 — a root of a polynomial, hence algebraic algebraic). (not rational, as it is not a ratio of polynomials).

7. Temperature in an industrial chamber increases linearly from 20°C at t = 0 to 80°C at t = 15 min. (1) Find T (t); (2) interpret the slope and intercept; (3) find T (8); (4) when will the chamber reach 60°C? (5) specify a physically reasonable domain.

[Source: Week 2, practical session, ex. 7]

Solution. (1) Slope m = $\frac{80 - 20}{15 - 0}$ = 4 ◦C/min, so T (t) = 20+4t. (2) The slope is the heating rate: +4 degrees per minute; the intercept 20 is the initial temperature. (3) T (8) = 20 + 32 = 52°C. (4) 20 + 4t = 60 ⇒ t = 10 min. (5) [0, 15] (or while the model remains linear).

8. For p(x) = x^3 − 4x determine: domain; intercepts with axes; symmetry; end behavior; points of discontinuity.

[Source: Week 2, practical session, ex. 8]

Solution. Domain R (polynomial). Intercepts: p(0) = 0; x^3 − 4x = x(x − 2)(x + 2) = 0 gives x = −2, 0, 2 — all on the x-axis through the origin. Symmetry: p(−x) = $-x^3$ +4x = −p(x) — odd. End behavior: the leading term x^3 with positive coefficient, so p(x) → +∞ as x → +∞ and p(x) →−∞ as x →−∞. The polynomial is defined everywhere: no points of discontinuity.

9. An electricity supplier charges C(E) = 0.10E for 0 ≤ E ≤ 200 and C(E) = 20 + 0.15(E − 200) for E > 200, in kWh. (1) Find C(150), C(200), C(350); (2) graph it; (3) is the model continuous at E = 200? (4) meaning of each slope; (5) one limitation.

[Source: Week 2, practical session, ex. 9]

Solution. (1) C(150) = 15; C(200) = 20; C(350) = 20 + 0.15 · 150 = 42.5 (all in $). (2) Two linear pieces: slope 0.10 up to 200, then 0.15 — a piecewise linear graph with a corner at (200, 20). (3) Yes: both formulas give 20 at E = 200, the pieces connect without a jump. (4) The slopes are marginal prices: 0.10$/kWh for the first 200 kWh, 0.15$/kWh beyond. (5) Limitation: the price jump exactly at 200 kWh incentivizes the consumer to stop just below the threshold; real tariffs change more smoothly or in blocks.

10. Find the inverse of f (x) = 4x − 7 and specify the domains; verify f (f^{-1}(x)) = x.

[Source: Week 2, practical session, ex. 10]

Solution. y = 4x − 7 ⇒ x = $\frac{y+7}{4}$, so f^{-1}(x) = $\frac{x+7}{4}$ with domain □ □ 4 and range R. Verification: f f^{-1}(x) = 4 · $\frac{x+7}{4}$ − 7 = x + 7 − 7 = x. 4 Both are straight lines with slopes 4 and $\frac{1}{4}$ — reciprocal slopes, mirror images with respect to y = x.

11. For each model, determine growth/decay and specify the initial value and growth/decay rate: A(t) = 500(1.08)^t, B(t) = 1200(0.85)^t, C(t) = 40$e^{0.03t}$, D(t) = 200$e^{-0.4t}$.

[Source: Week 2, practical session, ex. 11]

Solution. A: growth, initial 500, rate 1.08 (+8% per period). B: decay, initial 1200, rate 0.85 (−15%). C: growth, initial 40, continuous rate k = 0.03. D: decay, initial 200, continuous rate k = −0.4. The sign of k in e^{kt} or whether a > 1 in a^t

determines the direction.

12. Solve $2^{x+1} = 16$.

[Source: Week 2, practical session, ex. 12]

Solution. $16 = 2^4$, so $x + 1 = 4$ and $x = 3$. (Equivalently, take $\log_2$ of both sides.)

13. Battery discharge: $E(t) = E_0 e^{-0.025t}$, t in hours. (1) Fraction of remaining energy after 10 h? (2) When does 50% remain? (3) When does 10% remain? (4) Why is exactly zero unreachable?

[Source: Week 2, practical session, ex. 13]

Solution. (1) $E(10)/E_0 = e^{-0.25} \approx 0.7788$: about 77.9%. (2) $e^{-0.025t} = \frac{1}{2}$ $\ln 2$

$$E(t) = E_0 e^{-0.025t}$$

27.7 h. (3) $e^{-0.025t} = 0.1 \Rightarrow t = \frac{\ln 10}{0.025}$

$$E_0 e^{-0.025t} \approx 92.1 \text{ h. (4) The exponential is strictly positive}$$

for any finite t; it approaches zero only as $t \to \infty$ — first introduction to limits.

14. Population growth: $P(t) = 800e^{0.35t}$, t in hours. (1) Initial population? (2) After 5 h? (3) Doubling time? (4) Can the model remain realistic indefinitely?

[Source: Week 2, practical session, ex. 14]

Solution. (1) $P(0) = 800$. (2) $P(5) = 800e^{1.75} \approx 800 \cdot 5.755 \approx 4603$. (3) $T_d = \frac{\ln 2}{0.35} \approx 1.98$ h. (4) No: exponential growth ignores nutrient depletion and space; realistically, the model works only near the start of the culture.

3.3 Math I: Week 2 homework problems and complete solutions

1. Determine whether each relation defines y as a function of x: a. $y = 3x - 2$; b. $x^2 + y^2 = 9$; c. $y^2 = x + 1$; d. $x = y^3 - 2$.

[Source: Week 2, ISA, task 2]

Solution. a. Function. b. Not a function (circle: two y for each x). c. $y = \pm\sqrt{x + 1}$: two values for $x > -1$, not a function. d. Solve for $\sqrt[3]{x + 2}$; cube root is unique for each real y: $y = x$, so it's a function (passes vertical line test, though not horizontal).

2. For $f(x) = 2x^2 - 3x + 1$, calculate $f(0)$, $f(-2)$, $f(a)$, $f(a + h)$, $\frac{f(a+h)-f(a)}{h}$.

[Source: Week 2, ISA, task 3]

Solution. Same as exercise 2 of the practical session: 1; 15; $2a^2 - 3a + 1$; $2a^2 + 4ah + 2h^2 - 3a - 3h + 1$; difference quotient $= 4a + 2h - 3$.

$$(f(a+h) - f(a))/h = 4a + 2h - 3.$$

3. Find the natural domain: a. $f(x) = \frac{x-1}{q}$

$$sqrt((x+2)/(x-1))$$

c. $h(x) = \frac{}{x-4}$; d. $p(x) = \frac{x+2}{x-1}$.

[Source: Week 2, ISA, task 4]

Solution. a. $(-\infty, \infty)$. b. $(-\infty, \frac{5}{2}]$. c. Need $x - 1 \geq 0$ and $x \neq 4$: $[1, 4) \cup (4, \infty)$. d. Need $\frac{x+2}{x-1} \geq 0$ and $x \neq 1$: critical points -2 and 1; quotient non-negative on

$(-\infty, -2] \cup (1, \infty)$.

4. Find the range: $f(x) = x^2 + 2$, $g(x) = -|x - 1| + 4$, $h(x) = \sqrt{x + 3} - 2$.

[Source: Week 2, ISA, task 5]

Solution. $x^2 \geq 0 \Rightarrow f \geq 2$: range $[2, \infty)$. $|x - 1| \geq 0 \Rightarrow -|x - 1| \leq 0 \Rightarrow g \leq 4$: $(-\infty, 4]$. $\sqrt{x + 3} \geq 0 \Rightarrow h \geq -2$: $[-2, \infty)$. Each answer follows from the extreme value of the inner function.

5. Let $f(x) = x^2 - 1$, $g(x) = \sqrt{x + 2}$. Find $f + g$, $f - g$, fg, f/g and each new domain.

[Source: Week 2, ISA, task 6]

Solution. $(f \pm g)(x) = x^2 - 1 \pm \sqrt{x + 2}$; $(fg)(x) = (x^2 - 1)\sqrt{x + 2}$; $(f/g)(x) =$

$$(f + g)(x) = x^2 + \sqrt{x} + 1;\ (f - g)(x) = x^2 - \sqrt{x} - 3;\ (fg)(x) = (x^2 - 1)\sqrt{x} + 2;\ (f/g)(x) = (x^2 - 1)/\sqrt{x} + 2.$$

$\sqrt{x+2}$. Domains: $f + g$, $f - g$, fg: $x \geq -2$ (intersection of $\mathbb{R}$ and $[-2, \infty)$), i.e. $[-2, \infty)$. For f/g additionally $\sqrt{x + 2} = 0$, i.e. $x = -2$: $(-2, \infty)$.

6. Let $f(x) = 2x - 3$, $g(x) = x^2 + 1$. Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ f)(x)$, $(g \circ g)(x)$.

[Source: Week 2, ISA, assignment 7]

Solution. $(f \circ g)(x) = 2x^2 - 1$; $(g \circ f)(x) = 4x^2 - 12x + 10$; $(f \circ f)(x) = 2(2x - 3) - 3 = 4x - 9$; $(g \circ g)(x) = (x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$. The first two are different, so composition is non-commutative; the last two show that self-composition is perfectly valid.

7. Determine whether each function is even, odd, or of general form: $x^4 - 2x^2$; $x^3 - 4x$; $x^3 + x^2$; $|x| + 1$; $\frac{x^2}{x+1}$.

[Source: Week 2, ISA, assignment 8]

Solution. $x^4 - 2x^2$: only even powers — even. $x^3 - 4x$: only odd powers — odd. $x^3 + x^2$: mixed — general form. $|x| + 1$: even

$$x^2/(x+1)$$

(practical exercise 5). $\frac{}{x+1}$: numerator even, denominator changes sign: $\frac{x^2}{-x+1} = \pm\frac{x^2}{x+1}$ — general form.

√

8. Classify: $3x - 7$; $x^4 - 2x + 1$; $\frac{x^2 + 1}{x+1}$; 2^x; $\sin x$. $\frac{}{x^2+4}$;

$$(x^4 - 2x + 1)/(x + 1)$$

[Source: Week 2, ISA, assignment 9]

Solution. $3x - 7$: linear (polynomial). $x^4 - 2x + 1$: polynomial. $\frac{}{x^2+4}$: √ $x^2 + 1$: algebraic (root). 2^x: exponential — transcendental. $\sin x$: trigonometric — transcendental.

9. Linear temperature model from practical exercise 10 — all five parts.

[Source: Week 2, ISA, assignment 10]

Solution. Same model and answers as in practical exercise 7: $T(t) = 20 + 4t$; slope = heating rate 4 °C/min, intercept = initial 20 °C; $T(8) = 52$ °C; 60 °C at $t = 10$ min; domain $[0, 15]$.

10. For $p(x) = x^3 - 4x$ and $r(x) = \frac{x+1}{x-1}$ determine the domain, intercepts, symmetry, end behavior, points of discontinuity.

[Source: Week 2, ISA, assignment 12]

Solution. p: as in practical exercise 8 (domain R; intercepts −2, 0, 2; odd; ±∞ at ends; none). r: domain $x \neq 1$, i.e. $(-\infty, 1) \cup (1, \infty)$. Intercepts: $r(0) = -1$; numerator zero at $x = -1$. Symmetry: $r(-x) = \frac{-x+1}{-x-1} = \frac{x+1}{x-1} \neq \pm r(x)$ — general form. End behavior: $r(x) \to 1$ as $x \to \pm\infty$ (horizontal asymptote $y = 1$); $r(x) \to \pm\infty$ as $x \to 1$ (vertical asymptote $x = 1$). Undefined only at $x = 1$.

11. Piecewise electricity tariff from practical exercise 9 — all five parts.

[Source: Week 2, ISA, assignment 13]

Solution. Same answers as in practical exercise 9: $C(150) = 15$, $C(200) = 20$, $C(350) = 42.5$; continuous broken line graph; slopes — marginal prices; constraint — discontinuity in incentive at the 200 kWh boundary.

12. Convert 30°, 135°, 300° to radians and $\frac{\pi}{6}$, $5\frac{\pi}{4}$, $7\frac{\pi}{3}$ to degrees.

[Source: Week 2, ISA, assignment 14]

Solution. Multiply by π/180: $30^\circ = \frac{\pi}{6}$; $135^\circ = 3\frac{\pi}{4}$; $300^\circ = 5\frac{\pi}{3}$. Multiply by 180/π: $\frac{\pi}{6} = 30^\circ$; $5\frac{\pi}{4} = 225^\circ$; $7\frac{\pi}{3} = 420^\circ$.

13. Is it bijective? a. $f(x) = 3x - 5$, $x \in R$; b. $g(x) = x^2$, $x \in R$; c. $h(x) = x^2$, $x \geq 0$; d. $p(x) = \frac{1}{x}$, $x \neq 0$.

[Source: Week 2, ISA, assignment 15]

Solution. a. Yes: non-decreasing line passes horizontal line test. b. No: $g(-2) = g(2) = 4$. c. Yes: squaring is injective on $[0, \infty)$ (restriction restores horizontal line test). d. Yes: $\frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$.

14. Find inverses and their domains: a. $f(x) = 4x - 7$; b. $g(x) = \frac{x+3}{2}$; c. $h(x) = x^2 + 1$, $x \geq 0$; d. $p(x) = \frac{2x+3}{x-1}$, $x \neq 1$; verify by composition.

[Source: Week 2, ISA, assignment 16]

Solution. a. $f^{-1}(x) = \frac{x+7}{4}$, domain R (practical exercise 10). b. $y = \frac{x+3}{2} \Rightarrow$

$$y = sqrt(2)\,x + 1 \Rightarrow x = (y - 1)/sqrt(2)$$

$x = 2y - 3$, so $g^{-1}(x) = 2x - 3$, domain R. c. (positive root, since $x \geq 0$): $h^{-1}(x) = \sqrt{x - 1}$, domain $[1, \infty)$. d. $y = \frac{2x+3}{x-1} \Rightarrow y(x - 1) = 2x + 3 \Rightarrow x(y - 2) = y + 3 \Rightarrow x = \frac{y+3}{y-2}$: $p^{-1}(x) = \frac{x+3}{x-2}$, domain $x \neq 2$. Each satisfies $f(f^{-1}(x)) = x$ by construction.

15. Growth/decay classification for A, B, C, D (practical exercise 11).

[Source: Week 2, ISA, assignment 17]

Solution. Same: A growth ($a = 1.08$), B decay ($a = 0.85$), C growth ($k = 0.03$), D decay ($k = -0.4$).

16. Solve, giving exact answers first: a. $2^{x+1} = 16$; b. $5^{2x} = 125$; c. $e^{3x} = 7$; d. $100e^{-0.2t} = 20$.

[Source: Week 2, ISA, assignment 18]

Solution. a. $x + 1 = 4 \Rightarrow x = 3$. b. $2x = 3 \Rightarrow x = \frac{3}{2}$. c. $3x = \ln 7 \Rightarrow x = \frac{\ln 7}{3} \approx 0.6486$. d. $e^{-0.2t} = 0.2 \Rightarrow t = -\frac{\ln 0.2}{0.2} = 5 \ln 5 \approx 8.047$.

17. Battery discharge (exercise 13 from practical session) — all four parts.

[Source: Week 2, ISA, assignment 19]

Solution. Same: $e^{-0.25} \approx 77.9\%$ after 10 h; 50% at $\frac{\ln 2}{0.025} \approx 27.7$ h; 10% at $\frac{\ln 10}{0.025} \approx 92.1$ h; exponential never reaches zero in finite time.

18. Bacteria population (exercise 14 from practical session) — all four parts.

[Source: Week 2, ISA, assignment 20]

Solution. Same: $P_0 = 800$; $P(5) \approx 4603$; doubling time $\frac{\ln 2}{0.35} \approx 1.98$ h; purely exponential models fail in the long term (resources, space, waste).

4 Week 3 — Limits (Math I)

4.1 Math I: Formula catalog for week 3

4.1.1 Rates of change and two fundamental problems

Limits are introduced through two questions: how quickly a quantity changes (tangent/speed) and how function values behave near a point. Average speed on a shrinking interval determines instantaneous speed.

$$(f(x) - f(a))/(x - a) = m_{sec} \quad (51)$$

(slope of secant through (a, f (a))), (51) x − a

$$v_{ave} = (s(t) - s(a))/(t - a) \text{ (average speed, } v_{ave}\text{)}. \quad (59)$$

(average speed on [a, t]), (52) t − a f (a + h) − f (a)

$$\lim_{h \to 0} (f(a+h) - f(a))/h = f'(a) \text{ (instantaneous speed; defined as a limit)} \quad (53)$$

lim (instantaneous speed; defined as a limit) (53) h→0 h

4.1.2 Concept of limit, one-sided and infinite limits

Intuitive definition (OpenStax §2.2) and its one-sided variants — the operational heart of the week; two-sided limit exists precisely when the two one-sided limits coincide. $\lim_{x \to a}$ f (x) = L ⟺ all f (x) (x ≠ a) approach L as x → a, (54) $\lim_{x \to a-}$ f (x) = L, $\lim_{x \to a+}$ f (x) = L (left/right limits), (55) $\lim_{x \to a}$ f (x) = L ⟺ $\lim_{x \to a-}$ f (x) = $\lim_{x \to a+}$ f (x) = L (existence criterion), (56) $\lim_{x \to a-}$ f (x) = ±∞, $\lim_{x \to a+}$ f (x) = ±∞, $\lim_{x \to a}$ f (x) = ±∞ (infinite limits), (57) $\lim_{x \to a^-} (x - a)^n = -\infty$, $\lim_{x \to a^+} (x - a)^n = +\infty$ (n even); (58) $\lim_{x \to a^-} (x - a)^n = -\infty$, $\lim_{x \to a^+} (x - a)^n = +\infty$ (n odd) : x = a vert. asymptote. (59) The line x = a is a vertical asymptote of function f if at least one of the one-sided infinite limits (57) holds there.

4.1.3 Precise definition (§2.5, lecture emphasis)

The ε–δ definition turns "arbitrarily close" into quantitative distances; equivalent inequalities with absolute values — the working tool.

$$\lim_{x \to a} f(x) = L \iff \forall \varepsilon > 0 \; \exists \delta > 0 : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon, \quad (60)$$

$$|f(x) - L| < \varepsilon \iff L - \varepsilon < f(x) < L + \varepsilon, \quad (61)$$

$$0 < |x - a| < \delta \iff a - \delta < x < a + \delta \text{ and } x \neq a. \quad (62)$$

4.1.4 Two standard limits seen this week

sin x lim = 1 (lecture table 2.2: values 0.9983, 0.99998, 0.9999998 . . .), (63) x→0 x $_{x\to 0}$ sin 1 1 1 lim does not exist (oscillation), lim x = −∞, lim x = +∞. (64)

$$\lim_{x\to 0} (\sin x)/x = 1,\ \lim_{x\to 0} (\sin (1/x)) \text{ does not exist (oscillation)},\ \lim_{x\to 0-} (1/x) = -\infty,\ \lim_{x\to 0+} (1/x) = +\infty. \quad (64)$$

4.2 Math I: ISA for week 3 — questions and complete solutions

1. Study all problems, exercises, and examples presented in the lecture and practical session; be able to answer quickly and without errors.

[Source: Week 3, ISA, assignment 2]

Solution. This is a study instruction, not a calculation; the relevant material is fully solved in the lecture examples, in the practical session block below (exercises 2–8), and in the assigned textbook problems (following items).

2. For f (x) = x^3, estimate the slope of the tangent at x = 1, using m_{sec} = f(1+h)−f(1)

$$m_{sec} = 3 + 3h + h^2$$

from left and right to the same number?

[Source: Week 3, ISA, assignment 3]

Solution. Here f(1) = 1 and f(1 + h) = $(1 + h)^3 = 1 + 3h + 3h^2 + h^3$, so $m_{sec} = 3 + 3h + h^2$. Values: h = −0.1 ⇒ 2.71; −0.01 ⇒ 2.9701; −0.001 ⇒ 2.997; 0.001 ⇒ 3.003; 0.01 ⇒ 3.0301; 0.1 ⇒ 3.31. Both sides approach 3; indeed $\lim_{h\to 0}(3 + 3h + h^2) = 3$ — the exact slope of the tangent line (derivative of $3x^2$ at x = 1).

3. Solve problems 46–49 and 60–64 from "Chapter 1" (exercise list §2.2) of OpenStax Calculus 1.

[Source: Week 3, ISA, assignment 4]

Solution. These graph reading exercises are written out and fully solved below in §4.4; the numbering in the course sheet refers specifically to this exercise list.

4. A temperature controller outputs C(T) = 0 when T < 50 and C(T) = 10 when T ≥ 50 (volts). Determine $\lim_{T\to 50^-}$ C(T), $\lim_{T\to 50^+}$ C(T), $\lim_{T\to 50}$ C(T); interpret as a switching operation.

[Source: Week 3, ISA, assignment 5]

Solution. On the left, the controller reads the "off" branch: $\lim_{T\to 50^-}$ C(T) = 0. On the right — the "on": $\lim_{T\to 50^+}$ C(T) = 10. Since the one-sided limits are different, the two-sided limit does not exist (56). Interpretation: abrupt switching at T = 50 — the output jumps from 0 V to 10 V, which is exactly how a thermostat controls a heater.

$$s(t) = 2t^2 + 3t \text{ (m, s). (1)}$$

5. Position of a robot cart: (1) Average speed s(2+h)−s(2)

$$s(2+h)-s(2) = (2(2+h)^2 + 3(2+h)) - 14$$

(3) compute $\lim_{h\to 0}$; (4) interpret. h

$$s(3)-14 = (27-14)/(14.72-14)$$

[Source: Week 3, ISA, assignment 6]

Solution. (1) s(2) = 14. Values: h = 1: = 13; h = 0.1: =

$$s(2+h)-s(2)/h = (14.07(2+h)^2-14)/(h) = (14.07(4+4h+h^2)-14)/(h) = (56.28+56.28h+14.07h^2-14)/(h) = (42.28+56.28h+14.07h^2)/(h) = 42.28/h + 56.28 + 14.07h \ (9)$$

7.2; h = 0.01: $\frac{14.0702-14}{}$ = 7.02; h = −0.01: $\frac{13.9302-14}{}$ = 6.98. (2) s(2+h) = $2(2+h)^2$ +3(2+

$$s(2+h)-s(2) = 2(2+h)^2+3(2+h)-(2\cdot 2^2+3\cdot 2) = 8h+2h^2, \ (2)$$

h) = 8 + 8h + $2h^2$ + 6 + 3h = 14 + 11h + $2h^2$, therefore $\frac{s(2+h)-s(2)}{} = \frac{11h+2h2}{}$ = 11 + 2h. (3) h h $\lim_{h\to 0}$(11 + 2h) = 11 m/s. (4) The instantaneous speed of the cart at t = 2 s is 11 m/s; the average speeds approach it from both sides.

6. Two systems have outputs A(t) = $(1.08)^t$ and B(t) = $e^{0.03t}$. (1) Compute $\lim_{t\to\infty}$ A(t) and $\lim_{t\to\infty}$ B(t); (2) rewrite A(t) in the form e^{kt}; (3) which grows faster? (4) Compute $\lim_{t\to\infty \ A(t)}$ for B(t); (5) interpret.

[Source: Week 3, ISA, assignment 7]

Solution. (1) Both bases are greater than 1, so both limits equal +∞. (2) A(t) = $e^{t\ln 1.08}$, and ln 1.08 ≈ 0.0770, so A(t) = $e^{0.0770t}$. (3) A grows faster since its rate constant 0.0770 > 0.03. (4) $\frac{B(t)}{A(t)}$ = $e^{(0.03-0.0770)t}$ = $e^{-0.047t}$ → 0 as t →∞. (5) Although both diverge, their ratio approaches zero: B becomes negligible compared to A — the seed of "little o" comparison from week 4.

4.3 Math I: Week 3 practical session (engineering limits) — questions and complete solutions

Course requirements for this session: show mathematical procedure; include units; interpret in engineering terms; specify when the limit does not exist.

1. Solve problems 50–54 from the exercise list §2.2 of the textbook.

[Source: Week 3, practical session, exercise 1]

Solution. Written out and fully solved below in §4.4 (items 50–54).

2. One-sided limits: f(x) = 2x+1 when x < 2, f(x) = 7−x when x ≥ 2. Find $\lim_{x\to 2}$− f, $\lim_{x\to 2}$+ f, $\lim_{x\to 2}$ f, f(2); explain whether the two-sided limit exists.

[Source: Week 3, practical session, exercise 2]

Solution. Left branch: $\lim_{x\to 2}$−(2x + 1) = 5. Right branch: $\lim_{x\to 2}$+(7 − x) = 5. They are equal, so by (56) $\lim_{x\to 2}$ f(x) = 5. Note, f(2) = 7 − 2 = 5 as well — the function value equals the limit, so there is no discontinuity in the graph at x = 2.

$$V(T) = (2500/T)\ MB, \ T \geq 50$$

3. Calibration near a missing measurement: V(T) = mV, T ≠ 50 T −50 (T in °C). (1) Why is V(50) undefined? (2) Evaluate $\lim_{T\to 50}$ V(T); (3) how can the limit replace the missing calibration value?

[Source: Week 3, practical session, ex. 3]

Solution. (1) The formula divides by T − 50, which equals 0 when T = 50; the sensor model simply has no value there. (2) We factor the difference of squares: T^2 − 2500 = (T − 50)(T + 50), so when T ≠ 50, V(T) = T + 50, and $\lim_{T\to 50}$ V(T) = 100 mV. (3) The limit describes what the sensor would show at 50°C; filling the calibration table with 100 mV at this point is justified because the output

approaches this value from both sides—it's a "removable" behavior, not a jump.

4. Switching system: V(t) = 2t + 1 when t < 4 and V(t) = 12 − t when t ≥ 4 (V). Determine the one-sided limits, the two-sided limit, and V(4); is there a sharp voltage change at t = 4?

[Source: Week 3, practical session, ex. 4]

Solution. From the left: $\lim_{t\to4}-(2t + 1) = 9$ V. From the right: $\lim_{t\to4}+(12 − t) = 8$ V. V(4) = 12 − 4 = 8 V. The one-sided limits are different, so $\lim_{t\to4}$ V(t) does not exist: at t = 4, the voltage drops sharply from 9 V to 8 V. Engineering significance: a 1 V step can excite transients in the controlled device; when steps are harmful, input filters or soft-start logic are used.

5. Stress-strain: σ(ε) = 200000 ε when ε < 0.002 and σ(ε) = 400 + 10000(ε − 0.002) when ε ≥ 0.002 (σ in MPa; Hooke's region, then hardening). Find the one-sided limits at ε = 0.002; does the two-sided limit exist? Does the model predict a jump? Sketch the graph near the transition.

[Source: Week 3, practical session, ex. 5]

Solution. Elastic side: $\lim_{\varepsilon\to0.002}-$ 200000 ε = 200000(0.002) = 400 MPa.

$$\lim_{\varepsilon\to0.002^-}\sigma(\varepsilon) = 200000(0.002) = 400\ MPa;\ \lim_{\varepsilon\to0.002^+}\sigma(\varepsilon) = 400 + 10000(0.002 - 0.002) = 400\ MPa;\ \lim_{\varepsilon\to0.002}\sigma(\varepsilon) = 400\ MPa\ (exists).$$

Plastic side: 400 + 10000(ε − 0.002) = 400 MPa. Both equal 400 MPa, so the two-sided limit exists and equals 400 MPa; there is no jump—the piecewise model is continuous at the yield point, only the slope changes (from 200,000 to 10,000 MPa per unit strain). The graph is two straight line segments meeting at (0.002, 400) with a distinct corner: elastic loading, then a much shallower plastic branch.

6. Long-term response of an RC circuit: $V(t) = V_s(1 - e^{-t/(RC)})$ with V_s = 12 V, R = 10000 Ω, C = 100 μF. (1) Calculate $\lim_{t\to0}+$ V(t); (2) $\lim_{t\to\infty}$ V(t); (3) interpret both physically.

[Source: Week 3, practical session, ex. 6]

Solution. (1) At $t = 0^+$, the exponential equals $e^0 = 1$, so $V(0^+) = 12(1 − 1) = 0$ (2) As t →∞, $e^{-t/(RC)} \to 0$, V — initially uncharged capacitor. so $V \to V_s = 12$ V. (3) The capacitor charges from 0 V toward the supply voltage of 12 V, approaching it asymptotically; the time constant $\tau = RC = 10^4 \cdot 10^{-4} = 1$ s sets the pace (about 63% charge after 1 s, 99% after 5τ). The limiting language "$V \to V_s$ as t →∞" is precisely how asymptotic charging is formulated.

7. Terminal velocity: $v(t) = v_T\ (1 - e^{-kt})$, $v_T > 0$, k > 0. (1) v(0)? (2) $\lim_{t\to\infty}$ v(t)? (3) Why is this model more realistic than unlimited acceleration?

[Source: Week 3, practical session, ex. 7]

Solution. (1) $v(0) = v_T\ (1 − 1) = 0$: the object starts from rest. (2) $e^{-kt} \to 0$, so $v \to v_T$: the speed approaches the terminal value. (3) Air resistance increases with speed until it balances gravity; after that, the net force is zero, and speed can no longer increase—a falling body indeed approaches a finite limit, not accelerating forever.

8. Data transfer rate: $R(n) = {}^{100n}$ Mbps. (1) Calculate n+20 R(10), R(100), R(1000); (2) $\lim_{n\to\infty}$ R(n); (3) why does increasing n eventually give diminishing improvements? (4) maximum theoretical speed?

[Source: Week 3, practical session, ex. 8]

Solution. (1) R(10) ≈ 33.3, R(100) ≈ 83.3, R(1000) ≈ 98.0 Mbps. (2) Dividing by n: $R(n) = \frac{100}{1+20/n} \to 100$ Mbps. (3) Each additional unit of resource is divided by an increasingly large base, so the marginal benefit decreases — diminishing returns. (4) The horizontal asymptote 100 Mbps is the ceiling of the model: values approach it but do not exceed it.

4.4 Math I: assigned textbook exercises (OpenStax §2.2, exercises 46–54 and 60–64)

These graph reading exercises use four figures printed with the exercise set; the required values are read from the graphs as in the classroom. First the statements, then the solutions.

1. From the first graph y = f (x), solve: (46) $\lim_{x\to 10^-} f(x) = 0$; (47) $\lim_{x\to -2^+} f(x) = 3$; (48) $\lim_{x\to -8} f(x) = f(-8)$; (49) $\lim_{x\to 6} f(x) = 5$ — true or false?

[Source: OpenStax §2.2, ex. 46–49]

Solution. (46) True: as x → 10 from the left, the curve descends to the filled point (10, 0), so the left limit is 0. (47) True: slightly to the right of −2, the graph is at height 3 (filled point (−2, 3)). (48) False: near x = −8, the curve approaches an open circle at height −6, so $\lim_{x\to -8} f(x) = -6$, whereas f (−8) = −3 (filled point); the limit and value may not coincide. (49) False: from the left, the curve descends to an open circle (6, 2), so $\lim_{x\to 6^-} = 2$, whereas the right branch emerges from the filled point (6, 5), so $\lim_{x\to 6^+} = 5$; the one-sided limits are different ⇒ the two-sided limit does not exist, so it cannot equal 5.

2. For the piecewise linear graph (line y = x + 1 to the open point (1, 2); filled point (1, 1); then line y = 3 − x), find (50) $\lim_{x\to 1^-} f$; (51) $\lim_{x\to 1^+} f$; (52) $\lim_{x\to 1} f$; (53) $\lim_{x\to 2} f$; (54) f (1).

[Source: OpenStax §2.2, ex. 50–54]

Solution. (50) The left branch y = x + 1 → 2. (51) The right branch y = 3 − x → 2. (52) Both sides equal 2, so $\lim_{x\to 1} f(x) = 2$ — even though f (1) = 1; the limit ignores the point itself. (53) At x = 2, only the right branch exists: 3 − 2 = 1. (54) f (1) = 1 (filled point).

3. For the third graph (left: line descending to the open point (0, 1); right: curve rising from the filled point (0, −4) and crossing zero near x = 2), find (55) $\lim_{x\to 0^-} f$; (56) $\lim_{x\to 0^+} f$; (57) $\lim_{x\to 0} f$; (58) $\lim_{x\to 2} f$.

[Source: OpenStax §2.2, ex. 55–58]

Solution. (55) 1. (56) −4. (57) $1 \neq -4$, so the two-sided limit does not exist (56). (58) The right branch crosses the axis at x = 2, so the limit is 0.

4. For the fourth graph (three pieces: rising line to the open point (−2, 0); descending segment from the filled point (−2, 2) through the origin to (2, −2); rising line from the open point (2, 2)), find (59) $\lim_{x\to -2^-} f$; (60) $\lim_{x\to -2^+} f$; (61) $\lim_{x\to -2} f$; (62) $\lim_{x\to 2^-} f$; (63) $\lim_{x\to 2^+} f$; (64) $\lim_{x\to 2} f$.

[Source: OpenStax §2.2, ex. 59–64]

Solution. (59) 0. (60) 2. (61) $0 \neq 2$: does not exist. (62) −2. (63) 2. (64) $-2 \neq 2$: does not exist. Both junctions are actual jumps: no averaging can fix unequal one-sided limits.

5 Week 4 — Limits II: limit laws, standard limits, indeterminacies, asymptotics, Landau symbols (Math I)

Week 4 is the key week before Test 1 review: it turns the limit apparatus of week 3 into a working toolkit. Required reading is OpenStax §2.3 (Limit Laws), and week 4's notes by Prof. Neophytides summarize general limit properties, standard limits, indeterminacies, asymptotic equivalence, and Landau symbols O and o. This section lists each equation from both sources, then solves each exercise in the §2.3 set (exercises 83–130) and each worked question from the notes.

5.1 Formula catalog for week 4

5.1.1 General properties of limits (week 4 notes)

$\lim_{x\to a} f(x) = L$: $f(x)$ is arbitrarily close to L when x is close to a ($x \neq a$); (78) $\lim_{x\to a} f(x) = L \iff \lim_{x\to a-} f(x) = \lim_{x\to a+} f(x) = L$ (two-sided criterion), (79) the limit, when it exists, is unique.

5.1.2 Limit Laws (Theorem 2.5 OpenStax)

If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ (both exist) and c is a constant, then: Sum: $\lim_{x\to a}[f(x) + g(x)] = L + M$; Difference: $\lim_{x\to a}[f(x) - g(x)] = L - M$; (80) Constant Multiple: $\lim_{x\to a} c\,f(x) = cL$; Product: $\lim_{x\to a}[f(x)g(x)] = LM$; (81)

$$Power: \lim_{x\to a}[f(x)]^n = L^n \ (n\in N); (82)$$

Quotient: $\lim_{x\to a} f(x)/g(x) = L/M$ ($M \neq 0$); Power: $\lim_{x\to a}[f(x)]^n = L^n$ (82)

$$p_n = sqrt(n)$$

Root: $\lim_{x\to a}$ sqrt[n]{f(x)} = sqrt[n]{L} (n odd: any L; n even: $L \geq 0$); (83) Basic: $\lim_{x\to a} x = a$, $\lim_{x\to a} c = c$. (84)

$$\lim_{x\to a} p(x) = p(a)\ (85)$$

Polynomials/Rational: $\lim_{x\to a} p(x) = p(a)$, $\lim_{x\to a} q(x) = q(a)$ ($q(a) \neq 0$). (85) If $f(x) = g(x)$ for all $x \neq a$ near a, then $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$ — a "license" to cancel factors before substitution.

5.1.3 Squeeze Theorem (Theorem 2.7 OpenStax)

$f(x) \leq g(x) \leq h(x)$ near a ($x \neq a$), $\lim_{x\to a} f(x) = L = \lim_{x\to a} h(x) \Rightarrow \lim_{x\to a} g(x) = L$. (86)

5.1.4 Standard Limits and Derivable Trigonometric Limits (Week 4 summary)

$$\lim_{x\to 0} (e^x - 1)/x = 1,\ \lim_{x\to 0} (\sin x)/x = 1,\ \lim_{x\to 0} (1 - \cos x)/x^2 = 1,\ \lim_{x\to 0} (\ln(1 + x))/x = 1,$$
$$\lim_{x\to 0} x^n e^x = 0\ (n \in N),\ \lim_{x\to 0} (1 + x)^{1/x} = e,\ (87)$$

sin x 1 − cos x = 1 lim = 1, lim 2, lim = 1, (87)

$$lim\ x\to 0\ (e^x - 1)/x = 1,\ lim\ x\to 0\ \sin x/x = 1,\ lim\ x\to 0\ (1 - \cos x)/x^2 = 1/2,\ lim\ x\to 0\ \ln(1 + x)/x = 1,$$
$$lim\ x\to 0\ (1 + x)\text{^}(1/x) = e,\ lim\ x\to 0\ x^n/e^x = 0\ (n \in N).\ (87)$$

ln(1 + x) x^n

$$\lim_{x\to 0} (1 + x)^{1/x} = e,\ (88)$$

lim = 1, lim lim e_x = 0 (n ∈ N). (88)

$$\lim_{x\to 0} (\sin x)/x = 1,\ \lim_{x\to 0} (1-\cos x)/x^2 = 1,\ \lim_{x\to 0} (e^x - 1)/x = 1,\ \lim_{x\to 0} \ln(1+x)/x = 1,$$
$$\lim_{x\to 0} (1+x)^{1/x} = e,\ \lim_{x\to\infty} x^n/e^x = 0\ (n \in N).\ (88)$$

Angles in trigonometric limits must be in radians. Adaptation (87) by substitution, e.g., $\lim_{x\to 0} \sin 5x/x = 5 \lim_{x\to 0} \sin 5x/5x = 5$, $\lim_{\theta\to 0} (1-\cos\theta)/\theta = 0$ (conjugate). (89)

5.1.5 Indeterminate Forms (Week 4 summary)

0 ∞

$$0\cdot\infty,\ \infty-\infty,\ 0^0,\ 1\text{^}\infty,\ \infty^0$$

(90) The term "indeterminate form" does not mean the limit does not exist; it means additional analysis is required. The same initial form can resolve differently:

$$\lim_{x\to\infty} x/x^2,\ \lim_{x\to\infty} x^2/x,\ \lim_{x\to\infty} x/x\ (90)$$

lim x = 1, lim x = 0, lim x_2 does not exist (no finite), (91)

$$\lim_{x\to 0} (\sin x)/x = 1,\ \lim_{x\to 0} (1-\cos x)/x = 0,\ \lim_{x\to 0} (\sin(1/x)) \text{ does not exist (no finite)},\ (91)$$

while $\lim_{x\to 1} (x^2 - 1)/(x - 1) = 2$ after factoring. Resolution tools: factoring, rationalization, standard limits, asymptotic equivalence, substitution, algebraic rearrangement, squeeze theorem, Taylor expansions, and (later) L'Hôpital's rule.

5.1.6 Asymptotic Equivalence (Week 4 summary)

f (x)

$$f(x) \sim g(x)\ (x\to a) \iff \lim_{x\to a} f(x)/g(x) = 1,\ (92)\ f(x) \sim g(x) \iff f(x) = g(x)(1+\varepsilon(x)),\ \varepsilon(x)\to 0,$$
$$(93)\ f \sim g \iff f = g + o(g).\ (94)$$

$$x^2/2,\ 1-\cos x \sim x^2/2,\ e^x - 1 \sim x,\ (95)$$

As x → 0 : sin x ~ x, tan x ~ x, 2, (95)

$$(1+x)\text{^}\alpha - 1 \sim \alpha x\ (96)$$

ln(1 + x) ~ x, (α constant). (96) sin(0.01) Numerically ≈ 0.999983 illustrates "close but not equal." At infinity, the highest degree term dominates:

$$(2x^3 + x)/(5x^2 - 1) \sim (2x^3)/(5x^2) = (2/5)x\ (x\to\infty),\ (97)$$

Equivalences can be multiplied and divided, but not added in general.

5.1.7 Landau Symbols: Big O and Little o (Week 4 summary)

$$f(x) = O(g(x))\ (x\to a) \iff |f(x)/g(x)| \text{ remains bounded near } a,\ (98)\ f(x) = o(g(x))\ (x\to a) \iff$$
$$\lim_{x\to a} f(x)/g(x) = 0,\ (99)\ f = o(g) \Rightarrow f = O(g);\ f = O(g)\ ?\ f = o(g).\ (100)\ x2 = o(x),\ x3 = o(x2)$$

(x → 0); (101)

$$x = o(x^2),\ x^{10} = o(e^x)\ (x\to\infty).\ (102)$$

(x →∞). (102)

$$3x = O(x) \text{ but } 3x \neq o(x)\ (x\to 0) \text{ since } (3x)/x = 3.\ (103)\ x^2/2\ e^x = 1 + x + o(x);\ e^x = 1 + x + x^2/2 +$$
$$o(x^2).\ \sin x = x + o(x)\ (x\to 0);\ (104)$$

sin x = x + o(x) (x → 0); (104)

$$2x^2 + 3x + 1 = 2x^2 + O(x) \quad (104)$$

(x →∞). (105) In words: big O means that f has at most the order of magnitude of g; little o means that f becomes negligible compared to g. Caution: f = O(g) is not an equation in the usual sense — O(g) denotes a whole class of functions.

5.2 OpenStax §2.3, check problems 2.11–2.20 — questions and complete solutions

1. Using the limit laws, compute $\lim_{x\to -3}(2x-1)\sqrt{x+4}$, indicating the law at each step.

[Source: OpenStax §2.3, check 2.11]

Solution. Both limits of the factors exist, so we apply the product law: lim(2x−1) = 2(−3)−1 = −7 (sum, difference, basic limits); lim √x + 4 = √−3 + 4 = 1 (root law). Product: −7 · 1 = −7.

2. Compute $\lim_{x\to -2}(3x^3 - 2x + 7)$.

[Source: Check 2.12]

Solution. Direct substitution is justified (85): 3(−8) − 2(−2)+7 = −24+4+7 = −13.

3. Compute $\lim_{x\to -3} \frac{x^2+4x+3}{(x^2-9)}$.

[Source: Check 2.13]

Solution. 0/0 form. Factor both: $\frac{(x+1)(x+3)}{(x+3)(x-3)} = \frac{x+1}{x-3}$ when x ≠ −3. Limit: $\frac{-1}{-3} \frac{-1}{-3} = \frac{-2}{-6} = \frac{1}{3}$.

4. Compute $\lim_{x\to 5} \frac{\sqrt{x-1}-2}{x-5}$.

[Source: Check 2.14]

Solution. 0/0 form. Multiply by the conjugate √x − 1 + 2:

$$\frac{\sqrt{x-1}-4}{x-5} = \frac{1}{\sqrt{(x-5)(x-1+2)}} = \frac{1}{\sqrt{x-1+2}} \quad (x \neq 5)$$

so the limit $\frac{1}{4+2} = \frac{1}{4}$. √

$$\lim_{x\to 5} (x-5)$$

5. Compute $\lim_{x\to -3} \frac{x+3}{x+2}$.

[Source: Check 2.15]

Solution. Simplify the complex fraction: $\frac{1}{x+2} + 1 = \frac{x+3}{x+2}$, so the expression equals $\frac{x+3}{(x+2)(x+3)} = \frac{1}{x+2}$ when x ≠ −3. Limit: $\frac{1}{-3+2} = -1$. □ □

6. Compute $\lim_{x\to 3} \frac{1}{x-3} - \frac{4}{}$.

$$1/(x-3) - 1/(x+1)$$

[Source: Check 2.16]

Solution. Factor $x^2 - 2x - 3 = (x-3)(x+1)$ and simplify:

$$\frac{1}{(x-3)} - \frac{4}{((x-3)(x+1))} = \frac{((x+1)-4)}{((x-3)(x+1))} = \frac{(x-3)}{((x-3)(x+1))} = \frac{1}{(x+1)} \quad (x \neq 3),$$

so the limit $^{1}_{4}$. (

$$x^2 - x \le 3$$

7. Piecewise check: f (x) = x > 3; compute the one-sided x + 4 limits at the junction.

[Source: Check 2.17]

Solution. On the left, the parabolic branch gives $\lim_{x\to3}-$ f (x) = 9; on the right, the straight line gives $\lim_{x\to3}+$(x + 4) = 7. Not equal, so $\lim_{x\to3}$ f (x) does not exist.

8. Compute limx→1 $^{x}_{(x-}$ $^{+2}_{1)}$2 (method of example 2.23).

[Source: Check 2.18]

Solution. Numerator → 3 (nonzero), denominator → 0^+ from both sides (square), so the ratio → +∞: infinite limit, and x = 1 is a vertical asymptote.

9. Using the squeeze theorem, compute $\lim_{x\to0}$ x^2 sin $^{1}_{x}$.

[Source: Check 2.19]

Solution. Since −1 ≤ sin $^{1}_{x}$ ≤ 1, multiplying by x^2 ≥ 0 gives $-x^2 \le x^2 \sin \frac{1}{x} \le x^2$. Both bounds → 0; by (86) the limit is 0.

10. Compute $\lim_{\theta\to0}$ $^{1-\cos\theta}_{\sin\theta}$.

[Source: Check 2.20]

Solution. Divide numerator and denominator by θ (legal since within the limit θ ≠ 0): $\frac{1-\cos\theta}{\sin\theta} = \frac{1-\cos\theta}{\theta}\cdot\frac{\theta}{\sin\theta}$ using $\frac{1-\cos\theta}{\theta} \to 0$ (example 2.25) and $\frac{\sin\theta}{\theta} \to 1$ (standard limit).

$$(1 - \cos\theta)/(a \sin\theta) = 0,\ 1$$

5.3 OpenStax §2.3, exercises 83–130 — questions and complete solutions

5.3.1 Using limit laws (83–86)

1. $\lim_{x\to0}(4x^2 - 2x + 3)$.

[Source: §2.3, ex. 83]

Solution. Substitution by (85): $4(0)^2 - 2(0) + 3 = 3$.

2. $\lim_{x\to1}$ $^{x3+3x2+5}_{4-7x}$.

[Source: §2.3, ex. 84]

Solution. Denominator at 1 is −3 ≠ 0, so quotient law and substitution give $\frac{1+3+5}{-3} = \frac{9}{-3} = -3$. −3 √

$$lim_{x\to9}\ sqrt(x^2 - 6x + 3).$$

3. $\lim_{x\to-2}$

[Source: §2.3, ex. 85]

Solution. The radicand is continuous at −2 and positive there; root law: √4 + 12 + 3 = √ 19.

4. $\lim_{x\to-1}(9x + 1)^2$.

[Source: §2.3, exercise 86]

Solution. Power law, then substitution: $(9(-1)+1)^2 = (-8)^2 = 64$.

5.3.2 Direct Substitution (87–92)

1. $\lim_{x\to 7} x^2$.
[Source: §2.3, exercise 87]

Solution. 49.

2. $\lim_{x\to -2}(4x^2 - 1)$.
[Source: §2.3, exercise 88]

Solution. $4(4) - 1 = 15$.

3. $\lim_{x\to 0} \frac{1}{1+\sin x}$.
[Source: §2.3, exercise 89]

Solution. Denominator $1 + \sin 0 = 1 \neq 0$: quotient law gives $\frac{1}{1} = 1$.

4. $\lim_{x\to 2} e^{2x-x2}$.
[Source: §2.3, exercise 90]

Solution. Exponent: $2(2) - 4 = 0$; $e^0 = 1$.

5. $\lim_{x\to 1} \frac{2+7}{x+6}^{x}$.
[Source: §2.3, exercise 91]

Solution. $\frac{2+7}{1+6} = 9_7$.

6. $\lim_{x\to 3} \ln e^{3x}$.
[Source: §2.3, exercise 92]

Solution. $\ln e^{3x} = 3x$ identically (logarithm and exponential are inverses), so the limit is $3 \cdot 3 = 9$.

5.3.3 Indeterminate Form 0/0(93–102)

1. $\lim_{x\to 4} \frac{x2-16}{x-4}$.
[Source: §2.3, exercise 93]

Solution. Form 0/0. Factor: $\frac{(x-4)(x+4)}{x-4} = x + 4$ when $x \neq 4$; limit = 8.

2. $\lim_{x\to 2} \frac{x - 2}{x^2 - 2x}$.
[Source: §2.3, exercise 94]

Solution. Factor denominator: $\frac{x - 2}{x(x - 2)} = \frac{1}{x}$ when $x \neq 2$; limit = 1_2.

3. $\lim_{x\to 6} \frac{3x - 18}{2x - 12}$.
[Source: §2.3, exercise 95]

Solution. $\frac{3(x - 6)}{2(x - 6)} = 3_2$ when $x \neq 6$; limit = 3_2.

4. $\lim_{h\to 0} \frac{(1+h)^2-1)}{h}$.
[Source: §2.3, exercise 96]

Solution. $\frac{1+2h+h2-1}{h} = \frac{h(2+h)}{h} = 2 + h$; limit = 2 (this is the derivative of x^2 at 1).

5. $\lim_{t\to 9} \frac{t-9}{\sqrt{t}-3}$.
[Source: §2.3, exercise 97]

Solution. Form 0/0. Multiply by conjugate of denominator: $\sqrt{}$ $\sqrt{}$

$$\lim_{t\to 9} (t-9)/(\sqrt{t}-3) = \lim_{t\to 9} (t-9)(\sqrt{t}+3)/((\sqrt{t}-3)(\sqrt{t}+3)) = \lim_{t\to 9} (t-9)(\sqrt{t}+3)/(t-9) = \lim_{t\to 9} (\sqrt{t}+3) = 6 \ (t\neq 9)$$

so the limit is 9 + 3 = 6.

$$\frac{a+\frac{1}{h}-\frac{1}{a}}{h}$$

6. $\lim_{h\to 0} \frac{a}{h}$, where a is a nonzero real constant.
[Source: §2.3, exercise 98]

Solution. Simplify numerator: $\frac{a-(a+h)}{-h} = \frac{-1}{a(a+h)}$

$$\lim_{h\to 0} (a^{a+h} - a^{a})/h, \text{ where a is a nonzero real constant.}$$

Limit: $-\frac{1}{a^2}$ (derivative of 1/x at a).

7. $\lim_{\theta\to\pi} \frac{\tan\theta}{\sin\theta}$.
[Source: §2.3, exercise 99]

Solution. When $\theta \neq \pi$ near π: $\frac{\tan\theta}{\sin\theta} = \sin\theta \cdot \cos\theta = \cos\theta$ (cancellation is valid since $\sin\theta \neq 0$ there). Limit: $\cos\pi = -1$.

8. $\lim_{x\to 1} \frac{x^3-1}{x^2-1}$.
[Source: §2.3, exercise 100]

Solution. Factor both: $\frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{x^2+x+1}{x+1}$; limit $= \frac{3}{2}$.

$$(x-1)(x+1)/(x+1)$$

9. $\lim_{x\to 1/2} \frac{2x^2+3x-2}{2x-1}$.
[Source: §2.3, exercise 101]

Solution. $2x^2 + 3x - 2 = (2x - 1)(x + 2)$; cancel to $x + 2$; limit $= \frac{1}{2} + 2 = 5\frac{}{2}$.

$$sqrt(x + 4) - 1$$

10. $\lim_{x\to -3}$. $\frac{}{x+3}$

$$sqrt(x + 4 + 1)$$

[Source: §2.3, exercise 102]

Solution. Conjugate: multiply by $\sqrt{x+4}+1$:

$$((x + 4) - 1)/sqrt(x + 4 + 1) = 1/((x + 3)(x + 4 + 1)), (x \neq -3);$$

limit $= \frac{1}{1+1} = 1\frac{}{2}$.

5.3.4 Forms K/0: One-sided Infinite Limits (103–106)

1. $\lim_{x\to -2^-} \frac{2x^2+7x-4}{}$

$$(x+2)/(x-2)$$

[Source: §2.3, exercise 103]

Solution. Factor: numerator = $(2x-1)(x+4)$, denominator = $(x+2)(x-1)$. At $x = -2$, numerator = $(-5)(2) = -10 \neq 0$, while denominator $\to 0$: form K/0. Sign near -2: take $x = -2.1$: $(x + 2) < 0$, $(x - 1) < 0$, so denominator is positive; numerator $\approx -10 < 0$. Ratio $\to -\infty$.

2. $\lim_{x\to -2^+} \frac{2x^2+7x-4}{(x+2)\sqrt{(x-2)}}$

[Source: §2.3, exercise 104]

Solution. Same numerator; now $(x + 2) > 0$, $(x - 1) < 0$: denominator is negative. Ratio $\to +\infty$.

3. $\lim_{x\to 1^-} \frac{2x^2+7x-4}{(x+2)\sqrt{(x-2)}}$

[Source: §2.3, exercise 105]

Solution. Numerator at 1: $(1)(5) = 5 > 0$. When $x \to 1^-$: $(x + 2) > 0$, $(x - 1) < 0$, denominator is negative: ratio $\to -\infty$.

4. $\lim_{x\to 1^+} \frac{2x^2+7x-4}{(x+2)\sqrt{(x-2)}}$

[Source: §2.3, ex. 106]

Solution. The denominator is positive to the right of 1: the ratio $\to +\infty$. (The two-sided limits at -2 and at 1 both do not exist; $x = -2$ and $x = 1$ are vertical asymptotes.)

5.3.5 Limit Laws with Given Limits (107–114)

Throughout: $\lim_{x\to 6} f(x) = 4$, $\lim_{x\to 6} g(x) = 9$, $\lim_{x\to 6} h(x) = 6$.

1. $\lim_{x\to 6} 2f(x)\, g(x)$.

[Source: §2.3, ex. 107]

Solution. Product, then constant factor: $2 \cdot 4 \cdot 9 = 72$.

2. $\lim_{x\to 6} \frac{f(x)}{g(x)-1}$.

[Source: §2.3, ex. 108]

Solution. Law of quotients: $\frac{9-1}{\lim_{x\to 6} f(x)} = 2$.

3. $\lim_{x\to 6} f(x) + \frac{1}{3} g(x)$.

[Source: §2.3, ex. 109]

Solution. Sum and constant factor: $4 + \frac{9}{3} = 7$. □ □3.

4. $\lim_{x\to 6} h(x)$

[Source: §2.3, ex. 110]

Solution. Law of powers: $6^3 = 216$. p

5. $\lim_{x\to 6} g(x) - f(x)$.

[Source: §2.3, ex. 111]

Solution. Difference, then root: $\sqrt{9 - 4} = \sqrt{5}$.

6. $\lim_{x\to 6} x\, h(x)$.

[Source: §2.3, ex. 112]

Solution. Law of product with lim x = 6: 6 · 6 = 36.

7. $\lim_{x\to 6}[(x + 1) f(x)]$.
[Source: §2.3, ex. 113]

Solution. (6 + 1) · 4 = 28. □ □

8. $\lim_{x\to 6} f(x)\, g(x) - h(x)$.
[Source: §2.3, ex. 114]

Solution. 4 · 9 − 6 = 30.

5.3.6 Piecewise Functions and Limits from Graphs (115–125)

(

$$x^2 \le 3$$

1. f (x) = x > 3: find a. $\lim_{x\to 3^-}$ f, b. $\lim_{x\to 3^+}$ f. x + 4
[Source: §2.3, ex. 115]

Solution. a. Parabolic branch: 9. b. Linear branch: 7. (Two-sided limit does NOT exist.) (

$$(x^3 - 1)/x \le 0$$

2. g(x) = x > 0: find a. $\lim_{x\to 0^-}$ g, b. $\lim_{x\to 0^+}$ g. 1

$$0^3 - 1 = -1.$$

[Source: §2.3, ex. 116]

Solution. a. Cubic branch: b. Constant branch: 1. (Two-sided does NOT exist.)

(

$$x^2 - 2x + 1,\ x < 2$$

3. h(x) = x ≥ 2: find a. $\lim_{x\to 2^-}$ h, b. $\lim_{x\to 2^+}$ h. 3 − x
[Source: §2.3, ex. 117]

Solution. a. 4 − 4 + 1 = 1. b. 3 − 2 = 1. Both sides agree: $\lim_{x\to 2} h(x) = 1 = h(2)$ — continuous there, despite the piecewise definition.

4. Let f and g be two depicted functions (V-shaped f (x) = |x +2|; piecewise g with linear left branch g(x) = −x − 6 and curvilinear right branch), calculate $\lim_{x\to -3^+}$(f (x) + g(x)).
[Source: §2.3, ex. 118]

Solution. From the graph: f (−3) = |−3+2| = 1 and on the curvilinear branch $g(-3^+) = 1$. Law of sum: 1 + 1 = 2.

5. $\lim_{x\to -3-}$ □ □

$$f(-3) - 3g(-3^-) = 1 - 3(-1) =$$

[Source: §2.3, ex. 119]

Solution. Left linear branch gives $g(-3^-) = -(-3) - 6 = -3$... reading the junction from the graph, the value of the left branch at −3 is −1; we take the graph-value $g(-3^-) = -1$: f (−3) − 3g(−3⁻) = 1 − 3(−1) = 4.

$$\lim_{x\to -3-} (f(x) - 3g(x))$$

6. $\lim_{x\to 0}$. 3

[Source: §2.3, ex. 120]

Solution. f (0) = |2| = 2; g(0) = −2 (the curve passes through (0, −2)). Product and quotient: $\frac{2\cdot(-2)}{3} = -\frac{4}{3}$. 3

$$7.\ \lim_{x\to 3}(2+g(x))$$

7. $\lim_{x\to -3\ f(x)}$.

$$(2+g)/(-3-2) + (-1)/(2+1)$$

[Source: §2.3, ex. 121]

Solution. Left: $\lim_{x\to -3\ f(x)} = 1$; right: $\lim_{x\to -3\ f(x)} = 3$. One-sided

$$f(-3) = 1/2$$

limits are different, so two-sided limit does not exist. □ □$_2$.

8. $\lim_{x\to 1}$ f (x)

[Source: §2.3, ex. 122]

Solution. f (1) = |3| = 3; law of powers: 9.

$$P_3 = (1)/(3);\ \text{canonical form: } 3.$$

9. $\lim_{x\to 1}$ f (x) − g(x).

[Source: §2.3, ex. 123]

Solution. From the graph, g(1) = −5 and f (1) = 3: difference 8; cube root: $\sqrt[3]{8}$ = 2.

10. $\lim_{x\to -7}$ x g(x).

[Source: §2.3, ex. 124]

Solution. At x = −7 the linear branch gives g(−7) = −(−7) − 6 = 1; product: (−7)(1) = −7.

11. $\lim_{x\to -9}$[x f (x) + 2 g(x)].

[Source: §2.3, ex. 125]

Solution. f (−9) = | − 7| = 7; g(−9) = 9 − 6 = 3; (−9)(7) + 2(3) = −63 + 6 = −57.

5.3.7 Squeeze Theorem and Applications (126–130)

T True or false: if $2x-1 \le g(x) \le x^2-2x+3$ near x = 2, then $\lim_{x\to 2} g(x) = 0$.

[Source: §2.3, ex. 126]

Solution. Incorrect. Both bounds at x = 2 equal 2(2)−1 = 3 = 4−4+3, so by the squeeze theorem $\lim_{x\to 2} g(x) = 3$, not 0. (The theorem gives exactly the common limit of the bounds.) T $\lim_{\theta\to 0} \theta^2 \cos\frac{1}{\theta}$.

[Source: §2.3, ex. 127]

Solution. $-1 \le \cos\frac{1}{\theta} \le 1$ gives $-\theta^2 \le \theta^2 \cos\frac{1}{\theta} \le \theta^2$; both bounds → 0; squeeze: limit = 0. (0, x rational

1. $\lim_{x\to 0}$ f (x), where f (x) = x irrational.

$$\lim_{x\to 0} f(x) = 0$$

[Source: §2.3, ex. 128]

Solution. For all x: $0 \le f(x) \le x^2$ (rational x gives 0; irrational x gives $x^2 \ge 0$). Since $0 \to 0$ and $x^2 \to 0$, the squeeze theorem forces $\lim_{x\to 0} f(x) = 0$ — even though f is wildly discontinuous at every point $x \neq 0$. ^{q} 10^{x-19}; compute $\lim_{r\to 0^+}$ E(r) and T Coulomb's law: E(r) = when q =

$$E(r) = frac\{q\}\{4\,\pi\,\varepsilon_0\,r^2\}$$

interpret; why right?

[Source: §2.3, ex. 129]

Solution. The numerator is a positive constant; as $r \to 0^+$ the denominator $\to 0^+$, so $E(r) \to +\infty$: the field magnitude grows without bound as the test point approaches the charge. Only $r \to 0^+$ makes sense, because r is distance and cannot be negative (and r = 0 is the location of the charge itself). T Density ρ = m/V gives V (ρ) = $^{m}_{\rho}$; when m = 8 kg compute $\lim_{\rho\to 0^+}$ V (ρ) and explain the physical meaning.

[Source: §2.3, ex. 130]

$^{8}_{\rho}$; as $\rho \to 0^+$ volume $\to +\infty$:

Solution. V (ρ) = an object of fixed mass but vanishing density must occupy infinite volume (the formal statement of an "ideal gas, thinning out"). The right-hand limit is the only physical option, since density is positive.

5.4 Solved examples from week 4's notes — questions and complete solutions

1. Show that $\lim_{x\to 1} {}^{x2-1}_{x-1}$ has the form 0/0, and compute it; compare with $\lim_{x\to 0} {}^{x}_{x}$, $\lim_{x\to 0} {}^{x}_{x^2}$, $\lim_{x\to 0} {}^{x^2}_{x}$.

[Source: Week 4 notes, indeterminacies]

Solution. Substituting x = 1 gives $^{0}_{0}$ — indeterminacy. Factoring: (x−1)(x+1) = x + 1 when $x \neq 1$, so the limit is 2. The three comparisons resolve x−1

$$lim_{x\to 1}\,(x^2-1)/(x-1),\ lim_{x\to 0}\,x^x,\ lim_{x\to 0}\,x^2,\ lim_{x\to 0}\,x^{x^2}$$

the same form 0/0 in three different outcomes: = 1 → 1; = x → 0; =

$$lim_{x\to 1}\,(x^2-1)/(x-1)$$

$^{1}_{x}$ $\to \pm\infty$ (no finite two-sided limit exists). This is why 0/0 carries no automatic value: the outcome depends on how the numerator and denominator approach zero.

2. Using the standard limit $\lim_{x\to 0} {}^{\sin x}_{x} = 1$, compute $\lim_{x\to 0} {}^{\sin 5x}_{x}$. x x

[Source: Week 4 notes, standard limits]

Solution. Let u = 5x (so u → 0 as x → 0): $^{\sin 5x}_{5x} = 5\cdot {}^{\sin 5x}_{5x} = 5\cdot {}^{\sin u}_{u} \to 5\cdot 1 = 5$. x 5x u

3. Using asymptotic equivalence, compute $\lim_{x\to 0} {}^{\sin 3x}_{5x}$ and $\lim_{x\to 0} {}^{1-\cos x}_{x^2}$.

$$sin3x \sim 3x$$

[Source: Week 4 notes, asymptotics]

Solution. Since $\sin 3x \sim 3x$ as $x \to 0$, $^{\sin 3x}_{5x} \sim {}^{3x}_{5x} = 3_5$, so the limit is 3_5. 5x Since $1 - \cos x \sim {}^{x^2}_{2}$, $1 - {}^{\cos x}_{x^2/2} = {}^{1}_{2}$, so the limit is 1_2 — reproducing

$$lim_{x\to 0}\,(sin^3 x)/(5x) = 1/5\ (16)$$

the standard limit (87) via equivalence.

4. Show that $3x^2 + 5x - 1 \sim 3x^2$ as $x \to \infty$, and compute $\lim_{x\to\infty} \frac{2x^3+x}{5x^2-1}$ via equivalences.

[Source: Week 4 notes, equivalence at infinity]

Solution. $\frac{3x^2+5x-1}{} = 1 + \frac{5}{}\frac{1}{}$

$$\lim_{x\to\infty} (3x^2 + 5x - 1)/(3x^2) = 1 + 0 - 0 = 1, \text{ equivalence established:}$$

the highest degree term governs long-term behavior. Then $\frac{2x^3+x}{}$

$$\lim_{x\to\infty} (x^3+x)/(3x^2-1)$$

$\frac{2 \cdot x \cdot 3}{5 \cdot x \cdot 2} = \frac{2}{5}$ x, which → +∞; more precisely, the ratio of the original fraction to $\frac{2}{5}$ x approaches 1.

5. Show that sin x = O(x) as x → 0, and that 3x = O(x), but 3x ≠ o(x).

[Source: Week 4 Notes, Landau symbols]

Solution. The inequality | sin x| ≤ |x| holds for all real x (a standard geometric fact), which precisely means

$$|\sin x| \le 3|x|$$

that $\frac{}{|x|}$: sin x = O(x). For 3x: = 3 is bounded, so |x| 3x = O(x); but $\frac{3 \cdot x}{x} = 3 \not\to 0$, therefore 3x ≠ o(x). This is the most vivid illustration of the difference between O and o: big O allows any bounded ratio, little o requires the ratio to approach zero.

6. Verify $x^2 = o(x)$ as $x \to 0$ and $x = o(x^2)$, $x^{10} = o(e^x)$ as $x \to \infty$.

$$x^2 - 2x + x = o(x) \text{ near zero.}$$

[Source: Week 4 Notes, Landau symbols]

Solution. = x → 0: At infinity =

$$x^2 = o(x) \text{ as } x \to 0. \ (16)$$

$\frac{1}{}$ o(x^2); $\frac{x10}{}$ 0 by the standard limit $\lim_{x\to\infty} \frac{xn}{}$ → 0: x = and → = x e^x e^x 0 (88): exponential growth eventually dominates any fixed power.

7. Write $\sin x \sim x$ and $e^x = 1 + x + \frac{x^2}{2} + o(x^2)$ in terms of little o. Why is f (x) = O(g(x)) not an ordinary equation?

[Source: Week 4 Notes, relations]

Solution. By (94), f ~ g if and only if f = g + o(g): thus sin x ~ x is rewritten as sin x = x + o(x) — the error is negligible compared to x. For the exponential e^x = 1 + x + o(x) holds because $\frac{ex-1-x}{}$ → 0, x and keeping one more term gives e^x = 1+x+ $\frac{x^2}{2}$ +o(x^2). Finally, f = O(g) is not an equation because O(g) denotes an entire class of functions bounded by constant multiples of g; the notation $2x^2$ +3x +1 = $2x^2$ + O(x) means precisely that the remainder 3x + 1 grows no faster than a constant multiple of x.

6 Textbook supplements — additional exercises for required reading with full solutions

6.1 OpenStax §1.2 (Basic Classes of Functions) — selected exercises solved

1. For each pair of points, find the slope of the line through them and say whether the line is increasing, decreasing, horizontal, or vertical: (59) (−2, 4) and (1, 1); (61) (3, 5) and (−1, 2); (63) (2, 3) and (5, 7); (65) (2, 4) and (1, 4); (66) (1, 4) and (1, 0).

[Source: OpenStax §1.2, exercises 59–66]

Solution. Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$. (59) $m = \frac{1-4}{-1-2} = \frac{-3}{-3} = 1$: increasing. (61) $m = \frac{7-3}{5-2} = \frac{4}{3}$: increasing. (63) $m = \frac{4-4}{1-2} = 0$: horizontal. (66) $\Delta x = 0$: vertical, slope is undefined.

2. Write the equation in slope-intercept form: (67) slope −6, through (1, 3); (70) slope $\frac{2}{5}$, x-intercept 8; (71) through (2, 1) and (−2, −1); (73) intercepts: x at 5 and y at −3.

[Source: OpenStax §1.2, exercises 67–73]

Solution. (67) $y - 3 = -6(x - 1) \Rightarrow y = -6x + 9$. (70) x-intercept at 8 means the point (8, 0): $y - 0 = \frac{2}{5}(x - 8) \Rightarrow y = \frac{2}{5}x - \frac{16}{5}$. (71) $m = \frac{1-(-1)}{2-(-2)} = \frac{2}{4} = \frac{1}{2}$; $y - 1 = \frac{1}{2}(x - 2) \Rightarrow y = \frac{1}{2}x$. (73) through (5, 0) and (0, −3): $m = \frac{-3-0}{0-5} = \frac{3}{5}$; $y = \frac{3}{5}x - 3$.

3. (75–76) For each linear equation, specify the slope m and intercept b: $y = 2x - 3$; $y = -\frac{1}{7}x + 1$.

[Source: OpenStax §1.2, exercises 75–76]

Solution. (75) $m = 2$, $b = -3$. (76) $m = -\frac{1}{7}$, $b = 1$.

4. (81–82) Rewrite with slope and intercept: $2x + 3y = 6$; $6x - 5y + 15 = 0$.

[Source: OpenStax §1.2, exercises 81–82]

Solution. (81) $y = 2 - \frac{2}{3}x$: $m = -\frac{2}{3}$, $b = 2$. (82) $y = \frac{6}{5}x + 3$: $m = \frac{6}{5}$, $b = 3$.

5. (83–86) For each polynomial: degree; zeros; y-intercept; behavior at the ends; parity. a. $f(x) = 2x^2 - 3x - 5$; c. $f(x) = x^3 + 3x^2 - x - 3$.

[Source: OpenStax §1.2, exercises 83–86]

Solution. a. Degree 2; $2x^2 - 3x - 5 = (2x - 5)(x + 1)$: zeros $\frac{5}{2}$ and -1; y-intercept -5; branches upward (leading coefficient $2 > 0$), so $f \to +\infty$ at both ends; neither even nor odd (due to the odd term $-3x$ and the constant). c. Degree 3; testing $x = -1$: $-1 + 3 + 1 - 3 = 0$, so $(x + 1)$ is a factor: $x^3 + 3x^2 - x - 3 = (x + 1)(x^2 - 1) = (x + 1)(x - 1)(x + 1) = (x - 1)(x + 1)^2$: zeros 1 (simple) and -1 (double); y-intercept -3; cubic with positive leading coefficient: $f \to +\infty$ as $x \to +\infty$, $f \to -\infty$ as $x \to -\infty$; neither even nor odd.

6. (88–91) Describe the graph of g based on the basic graph: (88) $g(x) = x^2 - 1$; (89) $g(x) = (x + 3)^2 + 1$; (90) $g(x) = \sqrt{x + 2}$; (91) $g(x) = -\sqrt{x} - 1$.

[Source: OpenStax §1.2, exercises 88–91]

Solution. (88) Parabola $y = x^2$, shifted down by 1. (89) Parabola, shifted left by 3 and up by 1 (vertex $(-3, 1)$). (90) Square root graph, shifted left by 2 (domain

[−2, ∞)). (91) Square root graph, reflected across the x-axis and shifted down by 1 (domain [0, ∞), values ≤−1). (

$$f(x) = \{4x + 3,\ x \le 0;\ (9)\}$$

7. (94) f (x) = x > 0: calculate f (−3), f (0), f (2); sketch.

$$-x + 1$$

[Source: OpenStax §1.2, exercise 94]

Solution. f (−3) = 4(−3) + 3 = −9 (left branch); f (0) = 3; f (2) = −2 + 1 = −1 (right branch). Graph: increasing line to (0, 3), then decreasing line from open point (0, 1) — jump discontinuity at 0. (

$$f(x) = (x-3)/(x-2),\ x \neq 2$$

8. (97) g(x) = x = 2: calculate g(0), g(−4), g(2). 4,

[Source: OpenStax §1.2, exercise 97]

Solution. g(0) = $\frac{3}{-2} = -\frac{3}{2}$; g(−4) = $\frac{3}{-6} = -\frac{1}{2}$; g(2) = 4 by special definition (formula undefined there — hence piecewise value given). $\sqrt[3]{x}$ —

9. True or false (98–103): (98) f (x) = $\frac{4x+1}{7x-2}$ is transcendental; (99) g(x) = is an odd root function; (100) logarithmic function is algebraic; (101) f (x) = x^b with real constant b is an exponential function; (102) domain of even-degree root is all real numbers; (103) company bought equipment for $20,500, after 3 years worth $12,300 — find V (t) and interpret.

[Source: OpenStax §1.2, exercises 98–103]

Solution. (98) False: ratio of polynomials is rational, hence algebraic, not transcendental. (99) True: cube roots defined for all reals and function is odd. (100) False: x^b is a power function; logarithms are transcendental. (101) False: exponential has variable in exponent, b^x. (102) False: even-degree roots require non-negative radicand, so domain excludes negatives (odd roots accept all reals). (103) Linear depreciation: V (t) = 20500 + $\frac{12300-20500}{3}$ t = 3 20500 − $\frac{8200}{3}$ t ≈ 20500 − 2733.33t. V (0) = 20500 — purchase price; crosses t-axis at t ≈ 7.5 years — zero value moment; V (5) = 20500 − $\frac{41000}{3}$ ≈ $6833. 3

6.2 OpenStax §1.5 (Exponential and Logarithmic Functions) — selected exercises solved

1. (239–245) Sketch; specify domain, range, horizontal asymptote: a. f (x) = $e^x + 2$; b. f (x) = -2^x; c. f (x) = $1 - 2^{-x}$; d. f (x) = $e^{-x} - 1$.

[Source: OpenStax §1.5, exercises 239–245]

Solution. a. e^x, shifted up by 2: domain R, range (2, ∞), asymptote y = 2. b. Reflection across x-axis: domain R, range (−∞, 0), asymptote y = 0. c. As x → +∞, $2^{-x} \to 0$, so f → 1 from below; as x →−∞, $2^{-x} \to \infty$, so f →−∞: increasing, range (−∞, 1), asymptote y = 1. d. Mirror of e^x (decay right): domain R, range (−1, ∞), asymptote y = −1.

2. (246–253) Rewrite in equivalent exponential form: a. $\log_3 81 = 4$; b. $\log_8 2 = \frac{1}{3}$; c. $\log_5 1 = 0$; d. $\ln \frac{1}{e^3} = -3$.

[Source: OpenStax §1.5, exercises 246–253]

Solution. By (39): a. $3^4 = 81$. b. $8^{1/3} = 2$. c. $5^0 = 1$. d. $e^{-3} = \frac{1}{e^3}$.

3. (254–262) Rewrite in equivalent logarithmic form: a. $2^3 = 8$;

$$log_{64}\, 4 = 1/3$$

b. $4^{-2} = {}_{16}{}^{1}$; c. $10^2 = 100$; d. $64 = 4$; f. $9^y = 150.\ {}_{27}$; e. 3

$$log_{64}\, 4 = 1/3$$

[Source: OpenStax §1.5, exercises 254–262]

Solution. a. $\log_2 8 = 3$. b. $\log_{4\ 16} = -2$. c. $\log 100 = 2$. d. $\log_{1/3}$ means $\log_{64} 4 = {}^{1}{}_{3}$. f. $\log_9 150 = y$.

4. (264–266) Sketch; specify domain, range, vertical asymptote: a. f (x) = 3 + ln x; b. f (x) = ln(x − 1); c. f (x) = ln(−x).

[Source: OpenStax §1.5, exercises 264–266]

Solution. a. ln x, shifted up by 3: domain (0, ∞), range R, vertical asymptote x = 0. b. Shift right by 1: domain (1, ∞), range R, asymptote x = 1. c. Reflection across y-axis: domain (−∞, 0), range R, asymptote x = 0.

$$3HaueHHA\ R,\ ACHMIITOTAa\ zx = 0.$$

5. (270–275) Expand using logarithm laws: a. $\log x^4$; b. $\log {}^{9a3}{}_{b}$; c. ln a b; d. p $\log_5 125xy^3$.

[Source: OpenStax §1.5, exercises 270–275]

Solution. a. 4 log x. b. log 9 + 3 log a − log b = 2 log 3 + 3 log a − log b. c. ln a + ${}_{3}{}^{1}$ ln b. d. ${}^{1}{}_{5}\log_5(125xy^3) = 1_5\ (3 + \log_5 x + 3\log_5 y)$.

6. (276–283) Solve exactly: a. $5^x = 125$; b. $e^{3x} - 15 = 0$; c. $8^x = 4$; d. $4^{x+1} - 32 = 0$; e. $3^{x/14} = {}_{10}{}^{1}$; f. $10^x = 7.21$.

[Source: OpenStax §1.5, exercises 276–283]

Solution. a. x = 3 (since $125 = 5^3$). b. $3x = \ln 15$, $x = {}^{\ln 15} \approx 0.9027$. c. $2^{3x} = 2^2$, $3x = {}^{2}{}_{3}$. d. $4^{x+1} = 2^5 \Rightarrow 2^{2x+2} = 2^5 \Rightarrow x = 3\ {}_{14}{}^{x}$ ln 3 = − ln 10, $x = -\ {}^{14\ln 10}{}_{2}$. e. ≈−29.32. f. ln 3 x = log 7.21 ≈ 0.8580.

7. (284–291) Solve exactly, if possible: a. $\log_3 x = 0$; b. $\log_5 x = -2$; c. $\log_3(x+ 5) = 0$; d. $\log_2(2x - 7) = 0$; e. ln $\sqrt{x + 3} = 2$; f. $\log_6(x + 9) + \log_6 x = 2$; g. $\log_4(x + 2) - \log_4(x - 1) = 0$.

[Source: OpenStax §1.5, exercises 284–291]

Solution. a. $x = 3^0 = 1$. b. $x = 5^{-2} = {}_{25}{}^{1}$. c. $x + 5 = 3^0 = 1$, $x = -4$; check domain $x + 5 > 0$: $1 > 0$ OK, so $x = -4$ is valid ($\log_3 1 = 0$). d. $2x - 7 = 1$, $x = 4$. e. $\sqrt{x + 3} = e^2$, $x = e^4 - 3 \approx 51.60$. f. Combine: $\log_6 {}^{3}$

$$log_6(x(x + 9)) = 2$$

$x(x + 9) = 36 \Rightarrow x^2 + 9x - 36 = 0 \Rightarrow (x + 12)(x - 3) = 0$; domain requires $x > 0$, so $x = 3$. g. $\log_{4\ x\ -\ 1}{}^{x\ +2} = 0 \Rightarrow {}^{x\ +2}{}_{x\ -1} = 1 \Rightarrow x + 2 = x - 1$: impossible; no solutions.

8. (292–297) Change of base, approximation to four decimal places: a. $\log_5 47$; b. $\log_2 82$; c. $\log_{0.5} 211$; d. $\log_{0.2} 0.452$.

[Source: OpenStax §1.5, exercises 292–297]

Solution. By (41): a. ${}^{\ln\ 47}{}_{\ln\ 5} \approx 2.3922$. b. ${}^{\ln\ 82}{}_{\ln\ 2} \approx 6.3576$. c. ${}^{\ln\ 211}{}_{\ln\ 0.5} \approx -7.7181$. d. ${}^{\ln\ 0\ .452}{}_{\ln\ 0\ \ .2} \approx 0.3536$.

9. (307) A bacterial colony doubles in 12 hours, initially 1000. a. Using $Q = Q_0 e^{kt}$, find k; b. how long until 200,000?

[Source: OpenStax §1.5, exercise 307]

Solution. a. Doubling: $2 = e^{12k}$, so $k = \frac{\ln 2}{12} \approx 0.0578$ per hour. b. $200000 = 1000e^{kt}$: $e^{kt} = 200$, $t = \frac{\ln 200}{k} = \frac{12 \ln 200}{\ln 2} \approx 92.1$ hours.

10. (308) A rabbit population doubles every 6 months, initially 120. a. Using $P = P_0 a^t$, find a; b. when will it reach 3500?

[Source: OpenStax §1.5, exercise 308]

Solution. a. $a = 2$ for a 6-month period (or per year: with t in years $a = 4$; for 6 months $a = 2$). b. $3500 = 120 \cdot 2^t$ (with t in 6-month units): $2^t = \frac{175}{6} \approx 29.17$, $t = \log_2 29.17 = \frac{\ln 29.17}{\ln 2} \approx 4.87$ periods, i.e. about 2.4 years.

11. (309) The 1906 San Francisco earthquake had magnitude 8.3; the Japanese earthquake of the same year had magnitude 4.9. How much more intense was San Francisco (Richter scale)?

[Source: OpenStax §1.5, exercise 309]

Solution. Ratio of Richter intensities: $10^{8.3-4.9} = 10^{3.4} \approx 2512$: the San Francisco earthquake was about 2500 times more intense — a vivid demonstration of the logarithmic scale from (40).

6.3 OpenStax §2.2 — table exercises (42–45) solved

1. Estimate $\lim_{t \to 0^+} \frac{\cos t}{t}$ by table ($t = 0.1, 0.01, 0.001, 0.0001$). t

[Source: OpenStax §2.2, exercise 42]

Solution. Values: $t = 0.1$: $\frac{0.995004}{0.1} = 9.95$; $t = 0.01$: 99.95; $t = 0.001$: 999.95; $t = 0.1$ 0.0001: 9999.95. Values grow without bound: $\lim_{t \to 0^+} \frac{\cos t}{t} = +\infty$ (numerator $t \to 1$, while denominator $\to 0^+$ — K/0 form, as in (57)).

$$\lim_{t \to 0+} (\cos(t) - 1)/t$$

2. Estimate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by table.

$$(x^2-4)/(x-2)$$

[Source: OpenStax §2.2, exercise 43]

Solution. First simplify: $\frac{x^2 - 4}{x - 2} = \frac{(x-2)(x+2)}{x-2} = x + 2$ for $x \neq 2$. Table values: $x = 1.99$: $1.99 \cdot 3.99 = 7.9401$, $\frac{7.9401}{1} \approx 0.1259$; $x = 2.01$: $2.01 \cdot 4.01 = 8.0601$, ≈ 0.1241 — approaching $0.125 = \frac{1}{8}$. Algebra confirms the table: limit $\frac{1}{2 \cdot 4} = \frac{1}{8}$. T Guess from table, then verify by graph: $\lim_{\theta \to 0} \frac{\sin \theta}{\theta}$.

[Source: OpenStax §2.2, ex. 44]

Solution. Values: $\theta = 0.1$: $\sin(10\pi) = 0$; 0.01: $\sin(100\pi) = 0$; 0.001: $\sin(1000\pi) = 0$; 0.0001: 0. The table suggests 0 — but the graph oscillates between -1 and 1. $\frac{\pi}{2}$ gives $\sin \frac{\pi^2}{2}$ infinitely fast near 0 (for example, $\theta = \approx 1$). The limit does not exist; the table was misleading because the sample points $\frac{1}{n}$ were exactly zeros of sine. This is a cautionary exercise about the tabular method. T Guess from the table, then verify with the graph: $\lim_{\alpha \to 0^+} \frac{1}{\alpha} \cos \frac{\pi}{\alpha}$.

[Source: OpenStax §2.2, ex. 45]

Solution. When $\alpha = \frac{1}{n}$, $\frac{1}{\alpha} \cos(n\pi) = n(-1)^n$ — the values $\pm n$ grow in magnitude and : alternate in sign. The graph oscillates between ever larger positive and negative values: the limit does not exist (neither $+\infty$ nor $-\infty$ — the signs

alternate). The table at lucky points ($\alpha = \frac{1}{2n}$, gives $+2n \to +\infty$) would have been misleading in the opposite direction.